Fully device-independent model on quantum networks

Ming-Xing Luo

Information Security and National Computing Grid Laboratory, School of Information Science and Technology, Southwest Jiaotong University, Chengdu 610031, China

(Received 12 September 2020; accepted 24 February 2022; published 15 March 2022)

The Bell inequality can provide a useful witness for device-independent applications with quantum (or postquantum) eavesdroppers. This feature holds only for single entangled systems. Our goal is to explore a device-independent model for quantum networks. We first propose a Bell inequality to verify the genuinely multipartite nonlocality of connected quantum networks including cyclic networks and universal quantum computational resources for a measurement-based computation model. This is further used to construct a monogamy relation in a fully device-independent model with multisource quantum resources. It is finally applied for multiparty quantum key distribution, blind quantum computation, and quantum secret sharing. The present model can inspire various large-scale applications on quantum networks in a device-independent manner.

DOI: 10.1103/PhysRevResearch.4.013203

I. INTRODUCTION

The Bell theorem states that the statistics generated by local measurements on a two-spin entanglement cannot be generated by any classical local model under the locality and causality assumptions. This provides an experimental method for verifying the so-called nonlocality of entanglement [1,2]. To date, Bell theory has inspired many interesting applications in various areas, such as quantum information processing [3], quantum key distribution [4–8], and randomness amplification [9–11].

Single entangled systems have experimental constraints in large-scale applications because of the limited coherence time and transmission distance. This inspires distributed settings in terms of quantum networks using various independent entangled systems [12,13]. Compared with single entangled systems, there is great difficulty in characterizing multipartite correlations of quantum networks because of the independent assumption of sources and local joint measurement allowed for each party. Moreover, the participation of multiple parties in a general network may provide new opportunities for attackers. Thus, a natural problem is how to ensure secure information processing on general quantum networks. The main difficulty is that these multipartite quantum correlations can form nonconvex semialgebraic sets [14]. Several nonlinear Bell inequalities have recently been proposed for verifying the nonmultilocality of special quantum networks with the assumption of source independence [15–27]. However, so far there is no way to verify the multipartite nonlocality of general quantum networks, and no standard definition of genuinely multipartite nonlocality for quantum networks.

Published by the American Physical Society under the terms of the Creative Commons Attribution 4.0 International license. Further distribution of this work must maintain attribution to the author(s) and the published article's title, journal citation, and DOI.

In most secure tasks, such as quantum key distribution (QKD) [5], the trustworthiness of quantum devices according to certain specifications should be avoided in order to enable adversary (noise)-tolerant realizations [6]. These so-called device-independent scenarios only make use of the statistics of measurement outcomes [28–30]. Interestingly, the leaked information in the case of quantum eavesdroppers (or the key rate of QKD) may be characterized by the violation of a specific inequality [6–8], that is, the higher the violation is, the lower the information leakage for the outcomes of legal parties. Note that quantum devices may be correlated by an untrusted producer (adversary or eavesdropper) in a device-independent model [6]. This implies that the recent models [31,32] are not device independent because of the independent assumption of sources. The independence of sources in these secure scenarios cannot be guaranteed by experiment improvements. Another reason is that an untrusted party in network scenarios may correlate the shared sources locally in secure applications. Hence, the question of how to construct a device-independent model remains an open problem for general quantum networks.

Our goal in this work is to propose a device-independent model for secure information processing in general quantum networks against quantum (or postquantum) eavesdroppers. We first propose a new Bell inequality for verifying the genuinely multipartite nonlocality of quantum networks in the biseparable model [33]. Compared with a recent result for ring-shaped networks [34], the proposed inequality provides the first Bell test for verifying general cyclic networks. This further implies a new feature for characterizing the leaked information in device-independent tasks on quantum networks going beyond single entangled systems [6–8,28–30] or device-independent models [31,32]. It is then used to guarantee the security of various tasks, such as multipartite quantum key distribution, blind quantum computation with multiple servers, and quantum secret sharing using quantum

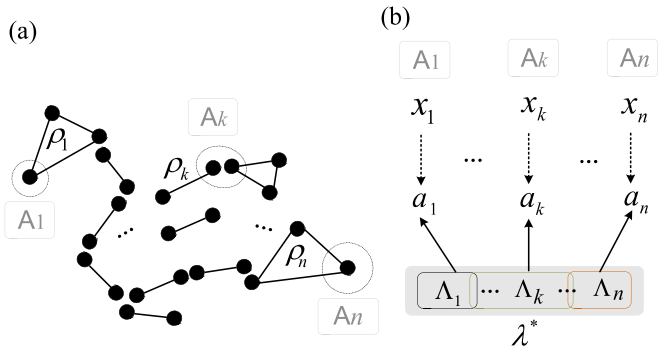


FIG. 1. Quantum networks. (a) An n-partite quantum network $\mathcal{N}_q$ consisting of m entangled states, $\otimes_{i=1}^m \rho_i$. (b) A directed acyclic graph (DAG) [35] of $\mathcal{N}_q$ consisting of m independent random variables $\lambda_1, \ldots, \lambda_m$. In device-independent applications, a quantum (or postquantum) eavesdropper can correlate shared variables λ_i into a new variable λ^*.

networks. These results can inspire interesting applications on large-scale quantum networks in a device-independent manner.

II. GENUINELY MULTIPARTITE NONLOCALITY OF QUANTUM NETWORKS

A. Bell inequalities

Consider an n-partite quantum network $\mathcal{N}_q$, as shown in Fig. 1(a), consisting of m entangled states $\otimes_{i=1}^m \rho_i$. $\mathcal{N}_q$ is schematically represented by a directed acyclic graph (DAG) [Fig. 1(b)] [35] with the same configuration as $\mathcal{N}_q$ using random variables $\lambda_1, \ldots, \lambda_m$. Different from the standard hidden variable model [1,2], a quantum (or postquantum) eavesdropper is able to correlate all distributed variables λ_i into a new variable λ^* [Fig. 1(b)]. Unfortunately, this local model is insufficient for characterizing device-independent applications including quantum secret sharing, where some sharers may be eavesdroppers. In this case, we consider the genuinely multipartite nonlocality of quantum networks in the biseparable model [33]. Let $I = \{s_1, \ldots, s_i\} \subset \{1, \ldots, n\}$ and $\bar{I} = \{j | j \notin I, 1 \leqslant j \leqslant n\}$ be a bipartition of $\{1, \ldots, n\}$. An n-partite state ρ on Hilbert space $\otimes_{i=1}^n \mathcal{H}_i$ is biseparable [33] if it has the following decomposition:

$$\rho = \sum_I p_I \varrho_{I,\bar{I}}, \tag{1}$$

where $\{p_I\}$ is a probability distribution, $\varrho_{I,\bar{I}}$ denotes a separable state on Hilbert space $\mathcal{H}_I \otimes \mathcal{H}_{\bar{I}}$, $\mathcal{H}_I = \otimes_{j\in I}\mathcal{H}_j$, and $\mathcal{H}_{\bar{I}} = \otimes_{j\in \bar{I}}\mathcal{H}_j$.

Let $M_{x_k(I)}$ denote a dichotomic observable of the observer A_k for input index $x_k(I)$, where x_k depends on the set I. Denote $\mathbf{M}_I = \prod_{k\in I} M_{x_k(I)}$ and $\mathbf{M}_{\bar{I}} = \prod_{k\in \bar{I}} M_{x_k(\bar{I})}$.

Theorem 1. Any biseparable state satisfies the following Bell inequality:

$$\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i \leqslant 2^n\sqrt{2} - 4\sqrt{2} + 2, \tag{2}$$

where $\mathcal{L}_i$ denotes the summation of multipartite Clauser-Horne-Shimony-Holt (CHSH)-type quantities, that is, $\mathcal{L}_i = \sum_{|I|=i} CH_I$ with $CH_I = (\mathbf{M}_I + \hat{\mathbf{M}}_I)\mathbf{M}_{\bar{I}} + (\mathbf{M}_I - \hat{\mathbf{M}}_I)\hat{\mathbf{M}}_{\bar{I}}$, $\mathbf{M}_S$ and $\hat{\mathbf{M}}_S$ are observables depending on the set $S = I$ or $\bar{I}$, and $\lfloor x \rfloor$ denotes the maximum integer no more than x. The largest bound for quantum networks is $2^n\sqrt{2} - 2\sqrt{2}$.

Proof. Consider an n-partite quantum network $\mathcal{N}_q$ [as shown in Fig. 1(a)] consisting of n parties $\mathsf{A}_1, \ldots, \mathsf{A}_n$. Suppose that the total state of $\mathcal{N}_q$ is in a biseparable state ρ on Hilbert space $\otimes_{i=1}^n \mathcal{H}_i$ [4], that is, it can be decomposed into

$$\rho = \sum_I p_I \varrho_{I,\bar{I}}, \tag{3}$$

where $I = \{s_1, \ldots, s_k\} \subset \{1, \ldots, n\}$ and $\bar{I} = \{j | j \notin I, 1 \leqslant j \leqslant n\}$ are a bipartition of $\{1, \ldots, n\}$, $\{p_I\}$ is a probability distribution, $\varrho_{I,\bar{I}}$ denotes a separable state on Hilbert space $\mathcal{H}_I \otimes \mathcal{H}_{\bar{I}}$, $\mathcal{H}_I = \otimes_{j\in I}\mathcal{H}_j$, and $\mathcal{H}_{\bar{I}} = \otimes_{j\in \bar{I}}\mathcal{H}_j$. Here, the summation is over all the bipartitions of $\{1, \ldots, n\}$.

Denote $M_{x_i}^{a_i}$ as positive operator-valued measurement (POVM) performed by A_i, where $M_{x_i}^{a_i}$ satisfies $\sum_{x_i=0,1} M_{x_i}^{a_i} = \mathbb{1}$ for $a_i \in \{-1, 1\}$, $i = 1, \ldots, n$. These measurement operators satisfy the commutativity condition $[M_{x_i}^{a_i}, M_{x_j}^{a_j}] = M_{x_i}^{a_i} M_{x_j}^{a_j} - M_{x_j}^{a_j} M_{x_i}^{a_i} = 0$ for any i and j with $i \neq j$. The binary output $a_i \in \{-1, 1\}$ is conditional on the input $x_i \in \{0, 1\}$. All parties perform local measurements on the state ρ. From Born's rule, the output statistics is characterized by the joint probability conditional on inputs, that is,

$$P(\mathbf{a}|\mathbf{x}) = \mathrm{tr}\big(M_{x_1}^{a_1} \otimes \cdots \otimes M_{x_n}^{a_n} \rho\big), \tag{4}$$

where $\mathbf{x} = x_1 \cdots x_n$, and $\mathbf{a} = a_1 \cdots a_n$. For each input x_i, define M_{x_i} as a dichotomic observable given by $M_{x_i} = M_{x_i}^{a_i=0} - M_{x_i}^{a_i=1}$, $i = 1, \ldots, n$. With the distribution of $P(\mathbf{a}|\mathbf{x})$, an n-partite correlation in terms of ρ is defined by

$$\begin{aligned} \langle M_{x_1} \cdots M_{x_n} \rangle &= \mathrm{tr}\big(M_{x_1} \otimes \cdots \otimes M_{x_n} \rho\big) \\ &= \sum_{\mathbf{a}} (-1)^{\sum_{i=1}^n (a_i+3)/2} P(\mathbf{a}|\mathbf{x}). \end{aligned} \tag{5}$$

In what follows, each party makes use of multiple observables. Especially, for each bipartition I and $\bar{I}$, define $M_{x_j(I)}$ as a dichotomic observable of A_j for the input x_j depending on I; that is, the input x_j depends on I. For simplicity, denote

$$\mathbf{M}_I = \textstyle\prod_{j\in I} M_{x_j(I)}, \quad \mathbf{M}_{\bar{I}} = \prod_{j\in \bar{I}} M_{x_j(\bar{I})}. \tag{6}$$

The left-hand side of inequality (2) is linear for all the measurement observables M_{x_j}. All the biseparable states consist a convex set of $\{\sum_{I,\bar{I}} p_{I,\bar{I}} \varrho_I \otimes \varrho_{\bar{I}}\}$. So, the maximal bound of $\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i$ is achievable at the vertex of $\{\sum_{I,\bar{I}} p_{I,\bar{I}} \varrho_I \otimes \varrho_{\bar{I}}\}$, that is, product states. Hence, it is sufficient to prove inequality (2) by using the product state $\rho = \varrho_I \otimes \varrho_{\bar{I}}$.

From inequality (2) CH_I is defined by

$$CH_I = (\mathbf{M}_I + \hat{\mathbf{M}}_I)\mathbf{M}_{\bar{I}} + (\mathbf{M}_I - \hat{\mathbf{M}}_I)\hat{\mathbf{M}}_{\bar{I}}, \tag{7}$$

which is a multipartite CHSH-type operator [36] or Mermin operator [37], where $|\mathbf{M}_I|, |\hat{\mathbf{M}}_I|, |\mathbf{M}_{\bar{I}}|, |\hat{\mathbf{M}}_{\bar{I}}| \leqslant 1$.

Now, for each bipartition I_0 and $\bar{I}_0$ (which may be different from the bipartition I and $\bar{I}$), there are two subcases for any product state $\rho = \varrho_I \otimes \varrho_{\bar{I}}$:

(i) The state ρ is separable in terms of the bipartition I_0 and $\bar{I}_0$. In this case, we can regard all the parties A_j with $j \in I$ as one party, and all the parties A_t with $t \in \bar{I}$ as the other. The quantity of CH_I in Eq. (7) is then regarded as a bipartite CHSH-type quantity [36]. It follows that

$$CH_{I_0} \leqslant 2, \tag{8}$$

which holds for any product state $\rho = \varrho_I \otimes \varrho_{\bar{I}}$.

(ii) The state ρ is entangled in terms of the bipartition I_0 and $\bar{I}_0$. In this case, there are two parties A_s and A_t with $s, t \in I$, or $\bar{I}$ such that they are entangled in the bipartition I_0 and $\bar{I}_0$, and $s \in I_0, t \in \bar{I}_0$. It follows that

$$CH_{I_0} \leqslant 2\sqrt{2}. \tag{9}$$

Here, the upper bound $2\sqrt{2}$ may be achievable. Take $\varrho_I = |\phi\rangle_{12}\langle\phi|$ and $\varrho_{\bar{I}} = |\phi\rangle_{34}\langle\phi|$ as an example, where $|\phi\rangle$ is an Einstein-Podolsky-Rosen (EPR) state. $I_0 = \{1, 3\}$ and $\bar{I}_0 = \{2, 4\}$. A_1 and A_2 share one EPR state. The maximal bound in inequality (9) can be followed from the CHSH test [36].

Moreover, for each product state of $\rho = \varrho_I \otimes \varrho_{\bar{I}}$ with the given bipartition I and $\bar{I}$, there is at least one term CH_I in the summation of $\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i$ satisfies that $CH_I \leqslant 2$. Combined with the linearity of $\mathcal{L}_i$, it follows that

$$\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i \leqslant 2(N-1)\sqrt{2} + 2 \tag{10}$$

for any product state ρ, where N denotes the total number of CH_I's.

In what follows, we need to evaluate N. For an even integer n, define m_i as the total number of CH_I's for a given i with $i \leqslant \frac{n}{2}$. From the standard combination theory [38] we obtain

$$\begin{aligned} m_i &= C(n, i), \quad 1 \leqslant i < \frac{n}{2}, \\ m_j &= \frac{1}{2}C(n, j), \quad j = \frac{n}{2}, \end{aligned} \tag{11}$$

where $C(n, i)$ denotes the combination number of choosing i balls from n different balls without choosing order. From Eq. (6), it follows that

$$\begin{aligned} N &= \sum_{i=1}^{n/2} m_i \\ &= \frac{1}{2}\sum_{i=1}^{n-1} C(n, i) \qquad (12) \\ &= \frac{1}{2}\sum_{i=0}^{n} C(n, i) - 1 \qquad (13) \\ &= 2^{n-1} - 1, \qquad (14) \end{aligned}$$

where equality (12) is obtained by using the equalities $C(n, i) = C(n, n-i)$ with $i = 1, \ldots, \frac{n}{2} - 1$. Equality (13) is from the equalities $C(n, 0) = C(n, n) = 1$. Equality (14) is from the equality $\sum_{i=0}^{n} C(n, i) = 2^n$.

For an odd integer n, similar to Eq. (11), using the standard combination theory we obtain

$$m_i = C(n, i), \quad 1 \leqslant i \leqslant \left\lfloor \frac{n}{2} \right\rfloor, \tag{15}$$

where $\lfloor x \rfloor$ denotes the maximal integer which is no more than x. Similar to Eqs. (11)–(14) it is easy to prove that

$$N = \sum_{i=1}^{\frac{n}{2}} m_i = 2^{n-1} - 1. \tag{16}$$

From Eqs. (10) and (14), and the linearity of $\mathcal{L}_i$, it yields that

$$\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i \leqslant 2^n\sqrt{2} - 4\sqrt{2} + 2. \tag{17}$$

For a general quantum network, $CH_I \leqslant 2\sqrt{2}$ for each bipartition I and $\bar{I}$. From Eqs. (10), (14), and (16), it follows that

$$\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i \leqslant (2^n - 2)\sqrt{2}. \tag{18}$$

The upper bound is achievable for some quantum networks, which will be proved in the next section. This completes the proof. ■

Remark 1. The present inequality (2) may be trivial if all observables of $\mathbf{M}_I$, $\hat{\mathbf{M}}_I$, $\mathbf{M}_{\bar{I}}$, and $\hat{\mathbf{M}}_{\bar{I}}$ are different for each bipartition I and $\bar{I}$, where each party has two inputs and two outputs for each bipartition I and $\bar{I}$. In this case, the present inequality (2) is a summation of different CHSH inequalities [36], where each party has $2N$ inputs and two outputs. However, it is not trivial when each party has less than $2N$ inputs, or only a few inputs. This is the main concern in what follows. Generally, the number of inputs will depend on the involved state or quantum network. We can prove that the total number of inputs is no more than eight. This will be followed from the proof for quantum networks in the next sections.

Example 1. Consider a triangle network $\mathcal{N}_q$ [13] shared by three parties, Alice, Bob, and Charlie. We get the following inequality:

$$\begin{aligned} &A_1B_1C_1 + A_2B_1C_1 + A_1B_2C_2 \\ &-A_2B_2C_2 + A_3B_3C_1 + A_3B_4C_1 \\ &+A_4B_3C_2 - A_4B_4C_2 + A_3B_5C_3 \\ &+A_3B_5C_4 + A_4B_6C_3 - A_4B_6C_4 \leqslant 4\sqrt{2} + 2, \end{aligned} \tag{19}$$

which holds for all biseparable states. A_i, B_j, and C_k are dichotomic observables of A_1, A_2, and A_3, respectively, $i, k = 1, \ldots, 4$; $j = 1, \ldots, 6$. It means that Alice and Charlie have four inputs while Bob has six inputs.

Example 2. Different from Example 1, we can get another inequality with fewer inputs as

$$\begin{aligned} &A_1B_1C_1 + A_2B_1C_1 + A_1B_2C_2 \\ &-A_2B_2C_2 + A_3B_3C_1 + A_3B_4C_1 \\ &+A_4B_3C_2 - A_4B_4C_2 + A_3B_1C_3 \\ &+A_3B_1C_4 + A_4B_2C_3 - A_4B_2C_4 \leqslant 4\sqrt{2} + 2, \end{aligned} \tag{20}$$

which holds for all biseparable states. It means that each party has four inputs.

Inequality (2) can be regarded as Svetlichny-type inequality [33] with at most eight inputs. The present inequality (2)

may reduce to a Svetlichny inequality with specific settings [33]. It is useful for verifying the genuinely multipartite nonlocality of connected quantum networks consisting of EPR states [2] and Greenberger-Horne-Zeilinger (GHZ) states [39] or noisy quantum networks (including two examples of triangle networks) [40]. The present nonlocality is stronger than the nonmultilocality [15–26] or the nonlocality [34,36,39,41].

B. Genuinely multipartite nonlocality of quantum networks with generalized EPR states

Let $\mathcal{N}_q$ be an n-partite connected quantum network shared by $\mathsf{A}_1, \ldots, \mathsf{A}_n$. Suppose that $\mathcal{N}_q$ consists of generalized EPR states $|\phi_1\rangle, \ldots, |\phi_m\rangle$, where $|\phi_i\rangle$ is given by

$$|\phi_i\rangle = \cos_i |00\rangle + \sin\theta_i |11\rangle, \tag{21}$$

$\theta_i \in (0, \frac{\pi}{4})$. Here, the connectedness of the quantum network means that for each pair of $\mathsf{A}_i, \mathsf{A}_j$, there is a chain-type subnetwork $\mathcal{N}_{i\to j} = \{\mathsf{A}_i, \mathsf{A}_{s_1}, \ldots, \mathsf{A}_{s_k}, \mathsf{A}_j\}$, where two adjacent parties share entangled states. The main goal is to prove the genuinely multipartite nonlocality of $\mathcal{N}_q$ with even n in this section. The other will be proved in the next section.

Theorem 2. There are local observables for each party such that the output statistics violates inequality (2) with an even integer $n \geqslant 4$; that is, $\mathcal{N}_q$ is genuinely multipartite nonlocal in the biseparable model.

Lemma 1. For an even $n \geqslant 4$, $\mathcal{N}_q$ can be locally transformed into a new network $\mathcal{N}'_q$ without classical communication such that all parties have odd particles, where $\mathcal{N}'_q$ consists of generalized EPR states and four-particle GHZ states.

Proof of Theorem 2. Let M_{x_i} be a dichotomic observable of A_i for the input $x_i \in \{0, 1\}$. In what follows, it is sufficient to prove that there are local observable for each party on $\mathcal{N}_q$ such that the quantum correlations generated by local measurements violate inequality (2).

From Lemma 1, after local transformations are performed, each party A_j is relabeled as B_j who has n_j number of particles. From Eq. (21), the total state of $\mathcal{N}_q$ is given by

$$\rho = \otimes_{i=1}^{m} |\phi_i\rangle\langle\phi_i|, \tag{22}$$

where $|\phi_i\rangle$ is an EPR state or four-particle GHZ state. Define ϖ_q as the quantum violation given by

$$\varpi_q = \sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i, \tag{23}$$

where $\mathcal{L}_i$'s are defined in inequality (2) associated with the state ρ in Eq. (22). It is sufficient to evaluate ϖ_q by maximizing all the multipartite quantities CH_I in inequality (2).

For each set $I = \{s_1, \ldots, s_k\}$, consider CH_I first. Note that $\mathcal{N}_q$ is connected. For each pair of parties B_i and B_j with $i \in I$ and $j \notin I$, there is one chain-shaped subnetwork $\mathcal{N}_{i\to j} \subset \mathcal{N}'_q$ connecting them. There is one longest chain-shaped subnetwork in these subnetworks $\{\mathcal{N}_{s_t\to\ell}\}$, that is, $\mathcal{N}_{s_t\to\ell} \subset \mathcal{N}'_q$ connecting the parties B_{s_t} and B_ℓ, where $s_t \in I$ and $\ell \notin I$. For simplicity, define $\mathcal{N}_{s_t\to\ell}$ as

$$\mathcal{N}_{s_t\to\ell} = \{\mathsf{B}_{s_t}, \mathsf{B}_{l_1}, \ldots, \mathsf{B}_{l_v}, \mathsf{B}_\ell\}, \tag{24}$$

where $l_1, \ldots, l_v \notin I$.

In what follows, define observables for each party as

$$M_{x_{s_t}(I)} = \cos\theta\sigma_z^{\otimes n_{s_t}} + \sin\theta\sigma_x^{\otimes n_{s_t}}, \tag{25}$$

$$\hat{M}_{x_{s_t}(I)} = \cos\theta\sigma_z^{\otimes n_{s_t}} - \sin\theta\sigma_x^{\otimes n_{s_t}}, \tag{26}$$

$$M_{x_j(I)} = \sigma_z^{\otimes n_j}, \hat{M}_{x_j(I)} = \sigma_x^{\otimes n_j}, \forall j \neq s_t, \tag{27}$$

where n_j denotes the number of particles owned by B_j, and $X^{\otimes k}$ denotes the tensor of k copies of X. It is easy to show that $M_{s_1}M_{s_1}^\dagger = \hat{M}_{s_1}\hat{M}_{s_1}^\dagger = \mathbb{1}$ because n_1 is an odd integer. Here, $x_j(I)$ means that the input x_j depends on the set I. With these observables, from Eqs. (6) and (25)–(27) it follows that

$$\begin{aligned}\langle \mathbf{M}_I\mathbf{M}_{\bar{I}}\rangle &= \mathrm{tr}\big(\big(\cos\theta\sigma_z^{\otimes n_{s_t}} + \sin\theta\sigma_x^{\otimes n_{s_t}}\big)\big(\otimes_{j\neq s_t}\sigma_z^{\otimes n_j}\big)\rho\big)\\ &= \cos\theta\,\mathrm{tr}\big(\big(\sigma_z^{\otimes n_{s_t}}\otimes_{j\neq s_t}\sigma_z^{\otimes n_j}\big)\big(\rho_{s_t}\otimes_{j\neq s_t}\rho_j\big)\big) \quad (28)\\ &= \cos\theta. \quad (29)\end{aligned}$$

In Eq. (28), ρ_j denotes the state owned by B_j, for $j \in \{1, \ldots, n\}$. Equation (28) follows from Lemma 1. Note that $\mathrm{tr}(\sigma_x \otimes \sigma_z|\phi_i\rangle\langle\phi_i|) = \mathrm{tr}(\sigma_z \otimes \sigma_x|\phi_i\rangle\langle\phi_i|) = 0$ for any generalized EPR state $|\phi_i\rangle$. Moreover, $\mathrm{tr}(M_1 \otimes M_2 \otimes M_3 \otimes M_4|\phi_i\rangle\langle\phi_i|) = 0$ for any generalized four-particle GHZ state $|\phi_i\rangle$ when $M_i \in \{\sigma_z, \sigma_x\}$ and M_i's are different. With these equalities, it follows from equality (22) that $\mathrm{tr}((\sigma_x^{\otimes n_{s_t}})(\otimes_{j\neq s_t}\sigma_z^{\otimes n_j})(\rho_{s_t}\otimes_{j\neq s_t}\rho_j))$. Moreover, we get equality (29) from the equalities of $\mathrm{tr}(\sigma_z \otimes \sigma_z|\phi_i\rangle\langle\phi_i|) = 1$ for any generalized EPR state $|\phi_i\rangle$, and $\mathrm{tr}(\sigma_z^{\otimes 4}|\varphi_i\rangle\langle\varphi_i|) = 1$ for any generalized four-particle GHZ state $|\varphi_i\rangle$.

Similarly, we get that

$$\langle \hat{\mathbf{M}}_I\mathbf{M}_{\bar{I}}\rangle = \cos\theta, \tag{30}$$

$$\langle \mathbf{M}_I\hat{\mathbf{M}}_{\bar{I}}\rangle = \langle \hat{\mathbf{M}}_I\hat{\mathbf{M}}_{\bar{I}}\rangle = \sin\theta\prod_{j=1}^{m}\sin 2\theta_j. \tag{31}$$

From Eqs. (7) and (29)–(31), it follows that

$$\begin{aligned}CH_I &= \mathrm{tr}(((\mathbf{M}_I + \hat{\mathbf{M}}_I)\mathbf{M}_{\bar{I}} + (\mathbf{M}_I - \hat{\mathbf{M}}_I)\hat{\mathbf{M}}_{\bar{I}})\rho)\\ &= 2\cos\theta + 2\sin\theta\prod_{j=1}^{m}\sin 2\theta_j\\ &= 2\sqrt{1 + \prod_{j=1}^{m}(\sin 2\theta_j)^2} \quad (32)\end{aligned}$$

when θ satisfies $\cos\theta = 1/\sqrt{1 + \prod_{j=1}^{m}(\sin 2\theta_j)^2}$.

Generally, for each set $I \subset \{1, \ldots, n\}$, from the assumption in inequality (2), each party B_j can choose proper observables depending on the input $x_j(I)$ such that the joint statistics satisfies inequality (32). The main reason is that one may choose a different input $x_j(I)$ depending on I. This implies that inequality (32) holds for all the multipartite quantities CH_I simultaneously. Hence, there exist observables for each

Algorithm 1. Transforming $\mathcal{N}_q$ with local unitary operations and auxiliary particles in the state $|0\rangle$.

Input: A connected graph $\mathcal{G}$ with n vertices, where n is an even integer.
Output: A connected graph $\mathcal{G}'$ with n vertices, where each vertex has an odd number of nodes.
S0 (**Initialization**) Each cyclic graph $\mathcal{G}$ is decomposed into a set of cyclic subgraphs $\{\mathcal{G}_1, \ldots, \mathcal{G}_k\}$, where $\mathcal{G}_i$ denotes a subgraph with one cycle in which any two adjacent parties share some EPR states. This can be efficiently completed.[a] Denote A_i's as the vertices (or parties in the network) that have an even number of nodes (or particles in the network). Denote B_j's as the vertices (or parties in the network) that have an odd number of nodes (or particles in the network), where the subindices i, j of A_i and B_j are different for convenience.
S1 If all vertices have an odd number of nodes, then output the graph $\mathcal{G}$.
S2 There are vertices which have an even number of nodes. For convenience, assume that in each subgraph $\mathcal{G}_i$ there are vertices which have an even number of nodes. Otherwise, delete the subgraphs without vertices which have an even number of nodes.

For ℓ from 1 to k, consider $\mathcal{G}_\ell$ as follows.
(1) For two adjacent vertexes A_i and A_j in $\mathcal{G}_\ell$, if both of them have an even number of nodes in $\mathcal{G}_\ell$, they can transform one connected edge into one hyperedge with four nodes by adding one new node for each vertex [38]. This can be completed by transforming one shared EPR state $|\phi_i\rangle|0\rangle_i|0\rangle_j$ into one four-particle GHZ state $|\hat{\phi}_i\rangle = \cos\theta_i|0000\rangle + \sin\theta_i|1111\rangle$ by using local controlled-NOT operations and one auxiliary particle in networks. After these operations are performed, both vertices have odd numbers of nodes. They are then relabeled as B_i and B_j.
(2) For two adjacent vertices A_i and A_j in $\mathcal{G}_\ell$, if one of them, A_i for example, has an even number of nodes in $\mathcal{G}_\ell$, there exist other vertices $\mathsf{A}_{s_1}, \ldots, \mathsf{A}_{s_t}$ which have even numbers of nodes. Choose one vertex A_{s_1} such that there are other vertices $\mathsf{B}_{j_1}, \ldots, \mathsf{B}_{j_\ell}$ satisfying that $\mathsf{A}_i, \mathsf{B}_{j_1}, \ldots, \mathsf{B}_{j_\ell}, \mathsf{A}_{s_1}$ consist of a chain-shaped connected subgraph. Now, each pair of adjacent vertices can transform one edge into one hyperedge with four nodes by adding one new node for each vertex. After the local operations are performed, these vertices have odd numbers of nodes, where one node is added for A_i and A_{s_1}, while two nodes are added for $\mathsf{B}_{j_1}, \ldots, \mathsf{B}_{j_\ell}$. A_i and A_{s_1} are then relabeled as B_i and B_{s_1}, respectively. These two steps can be iteratively performed for all vertices in $\mathcal{G}_\ell$.

[a]For i from 1 to n, find a new cycle $\mathcal{G}_j$ started from the vertex A_i such that $\mathcal{G}_j \in \{\mathcal{G}_1, \ldots, \mathcal{G}_{j-1}\}$. Otherwise, there is no new cycle. For each j, there is at least one new vertex which will be added in $\mathcal{G}_j$ going beyond $\mathcal{G}_1, \ldots, \mathcal{G}_{j-1}$. This procedure will stop if all the vertices will be included in some cycle. The total time complexity is $O(n^2)$, where each cycle has at most n vertices.

observer such that the quantum correlations satisfy

$$\begin{aligned}\varpi_q &= \sum_i \mathcal{L}_i \\ &= 2N\left(1 + \prod_{j=1}^{m}(\sin 2\theta_j)^2\right)^{1/2} \\ &> 2^n\sqrt{2} - 4\sqrt{2} + 2 \end{aligned} \tag{33}$$

if all θ_j's satisfy the following inequality:

$$\prod_{j=1}^{m} \sin 2\theta_j > (N^2 - 2(2-\sqrt{2})N + 3 - 2\sqrt{2})^{1/2}, \tag{34}$$

where $N = 2^{n-1} - 1$.

From Eq. (32), the maximal violation is achieved by setting $\theta_j = \frac{\pi}{4}$ for all j's. This completes the proof for even n. ■

C. Genuinely multipartite nonlocality of quantum networks consisting of generalized GHZ states

Let $\mathcal{N}_q$ be an n-partite connected quantum network consisting of $\mathsf{A}_1, \ldots, \mathsf{A}_n$. Suppose that $\mathcal{N}_q$ consists of generalized GHZ states $|\phi_1\rangle, \ldots, |\phi_m\rangle$, where $|\phi_i\rangle$ is given by

$$|\phi_i\rangle = \cos_i |0\rangle^{\otimes m_i} + \sin\theta_i |1\rangle^{\otimes m_i}, \tag{35}$$

$\theta_i \in (0, \frac{\pi}{4})$. The main goal in this section is to prove the genuinely multipartite nonlocality of $\mathcal{N}_q$ by using inequality (2).

Theorem 3. There are local observables for each party such that the output statistics violates inequality (2); that is, $\mathcal{N}_q$ is genuinely multipartite nonlocal in the biseparable model.

Proof. Note that m_i can be assumed to be an even integer for each i. Otherwise, one party can change it into a new GHZ state with an even number of particles by using local operations and auxiliary particles. With this assumption, we first prove that $\mathcal{N}_q$ is genuinely n-partite nonlocal for even n with $n \geqslant 3$. Similar to local transformations defined in Algorithm 1, $\mathcal{N}_q$ can be locally transformed into a new network $\mathcal{N}_q'$ without classical communication such that all parties have odd numbers of particles, where $\mathcal{N}_q'$ consists of generalized EPR states and GHZ states with even numbers of particles. The proof is similar to Algorithm 1. The rest of the proof of the genuinely multipartite nonlocality is similar to the procedure from Eqs. (23)–(33).

In what follows, we prove the result for odd n with $n \geqslant 3$. Similar to local transformations defined in Algorithm 1, $\mathcal{N}_q$ can be locally transformed into a new network $\mathcal{N}_q'$ without classical communication such that there are $n-1$ observers who have odd numbers of particles, where $\mathcal{N}_q'$ consists of generalized EPR states and GHZ states with even numbers of particles. The proof is similar to Algorithm 1 by considering a connected subnetwork consisting of $n-1$ parties.

There are three subcases to prove the genuinely multipartite nonlocality of $\mathcal{N}_q'$:

(S1) Consider the subnetwork consisting of $n-1$ parties $\mathsf{A}_1, \ldots, \mathsf{A}_{n-1}$. These parties can locally transform the shared network into another subnetwork using local unitary operations without the help of classical communication such that they have odd numbers of particles. After these local transformations, these parties are relabeled by $\mathsf{B}_1, \ldots, \mathsf{B}_{n-1}$. The rest of the proof is similar to Eqs. (22)–(33) when A_n has an odd number of particles. Otherwise, there are two kinds of multi-

partite CHSH-type quantities CH_I which will be distinguished from their inputs.

(S2) Consider CH_I with $n \notin I$. Similar to Eq. (24), there exists one longest chain-shaped network $\mathcal{N}_{s_t\to\ell} = \{\mathsf{B}_{s_t}, \mathsf{B}_{j_1}, \ldots, \mathsf{B}_{j_t}, \mathsf{B}_\ell\}$, where $j_1, \ldots, j_t, \ell \notin I$. For the party A_n, define its local observable as

$$M_{x_n(I)} = \sigma_z^{\otimes h_n},$$
$$\hat{M}_{x_n(I)} = \sigma_x^{\otimes h_n}, \tag{36}$$

where h_n denotes the number of particles owned by the party B_n. With this definition, inequality (26) holds for each $I \subset \{1, \ldots, n\}$ satisfying $n \notin I$. From the assumption in inequality (2) in the main text, i.e., $t_I, \hat{t}_I$ are different for any I, the party A_n can choose proper local observables for these postselective terms CH_I simultaneously. Hence, inequality (32) holds for all I's satisfying $n \notin I$. The maximal violation is achieved by using maximally entangled GHZ states.

(S3) Consider CH_I with $n \in I$. In this case, from the assumption of inequality (2), for each I, A_n and A_i, who share at least one GHZ state, can transform locally the shared states, such that A_n owns an odd number of particles after local operations. The observables of A_n are similar to those given in Eqs. (23) and (25). And then, there exists another set of observables for all parties such that inequality (32) holds for CH_I. Note that all CH_I's can be distinguished by all parties using proper inputs. This means that inequality (32) holds for all the terms CH_I. Consequently, inequality (32) holds for each odd n. The maximal violation is achieved by using maximally entangled GHZ states.

This completes the proof. ■

D. The numbers of the inputs

In this section, we show that there are eight inputs for each party involved in inequality (2) for verifying general networks.

Consider an n-partite connected quantum network $\mathcal{N}_q$ consisting of EPR states and GHZ states. For an even $n \geqslant 3$, from Lemma 1 $\mathcal{N}_q$ can be locally transformed into a new network $\mathcal{N}_q'$ satisfying that each party has an odd number of particles, where $\mathcal{N}_q'$ consists of generalized EPR states and GHZ states with even numbers of particles.

Similar to Eq. (24), for each $I \subset \{1, \ldots, n\}$, there is one largest chain-type subnetwork $\mathcal{N}_{s_t\to\ell} = \{\mathsf{B}_{s_t}, \mathsf{B}_{l_1}, \ldots, \mathsf{B}_{l_v}, \mathsf{B}_\ell\}$ connecting two parties B_{s_t} and B_ℓ with $s_t \in I$ and $l_1, \ldots, l_v, \ell \notin I$. The quantum network $\mathcal{N}_q'$ can be verified by using inequality (2) with four inputs. In fact, for each party B_j, there are two subcases:

(i) In the case $j = s_t$, there are two input indices $x_j(I) \in \{1, 2\}$ associated with the observables shown in Eqs. (25) and (26).

(ii) In the case $j \neq s_t$, there are two input indices $x_j(I) \in \{3, 4\}$ associated with the observables shown in Eq. (27).

Note that s_t depends on the longest chain-type subnetwork $\mathcal{N}_{s_t\to\ell}$, which may be relaxed as a general chain-type subnetwork without requirement of the length. With this relaxation, one can assume that $s_t = \min\{s_j | s_j \in I\}$ and $\ell = \min\{j | j \notin I\}$. Denote

$$\mathbf{M}_I = M_{x_{s_t}=1} \prod_{j\in I, j\neq s_t} M_{x_j=3},$$
$$\hat{\mathbf{M}}_I = M_{x_{s_t}=2} \prod_{j\in I, j\neq s_t} M_{x_j=4},$$
$$\mathbf{M}_{\bar{I}} = \prod_{j\in\bar{I}} M_{x_j=3},$$
$$\hat{\mathbf{M}}_{\bar{I}} = \prod_{j\in\bar{I}} M_{x_j=4}. \tag{37}$$

With these definitions, inequality (2) can be rewritten into

$$\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i \leqslant 2^n\sqrt{2} - 4\sqrt{2} + 2, \tag{38}$$

where $\mathcal{L}_i = \sum_{|I|=i} CH_I$ with $CH_I = (\mathbf{M}_I + \hat{\mathbf{M}}_I)\mathbf{M}_{\bar{I}} + (\mathbf{M}_I - \hat{\mathbf{M}}_I)\hat{\mathbf{M}}_{\bar{I}}$. Similar to Secs. II B and II C, we can prove that Theorems 2 and 3 hold for any n-partite connected quantum network consisting of generalized EPR states and GHZ states.

For an odd n, from the proof in Appendix C, the quantum network of $\mathcal{N}_q$ can be locally transformed into two new networks $\mathcal{N}_q''$ and $\mathcal{N}_q'''$ satisfying that each party in $\mathcal{N}_q''$ or $\mathcal{N}_q'''$ has an odd number of particles. For the network of $\mathcal{N}_q''$, the genuinely multipartite nonlocality can be verified by inequality (38) with four inputs. A similar result holds for $\mathcal{N}_q'''$. Note that for each party B_j, there are four inputs as shown in Eq. (37) if local states of B_j are the same as each other in both networks $\mathcal{N}_q''$ and $\mathcal{N}_q'''$. Otherwise, there are at most eight inputs associated with two sets of observables as shown in Eq. (37). For some special network, it may be reduced to six inputs, where there are two different observables for verifying $\mathcal{N}_q'''$ and $\mathcal{N}_q'''$ (see the following example). Note that the number of inputs for each party depends on the transformed network $\mathcal{N}_q''$ or $\mathcal{N}_q'''$, which is from the network configuration of $\mathcal{N}_q$. We cannot get the explicit form of inequality (38) in this case.

Continuing Example 1. Consider the triangle network consisting of three EPR states by using inequality (19). Assume that two parties of Alice, Bob, and Charlie share one generalized EPR state: $|\phi_i\rangle_{2i-1,2i} = \cos\theta_i|00\rangle + \sin\theta_i|11\rangle$ with $\theta_i \in (0, \pi)$ and $i = 1, 2$, and 3. The total system is denoted as $|\phi_1\rangle_{12}|\phi_2\rangle_{34}|\phi_3\rangle_{56}$. Alice owns particles 1 and 6. Bob owns particles 2 and 3 while Charlie owns particles 4 and 5. Here, $|\phi_1\rangle$ and $|\phi_3\rangle$ will be changed into four-particle GHZ states with local operations and auxiliary qubits. Define local observables as

$$A_1 = B_3 = C_3 = \cos\theta\sigma_z^{\otimes 3} + \sin\theta\sigma_x^{\otimes 3},$$
$$A_2 = B_4 = C_4 = \cos\theta\sigma_z^{\otimes 3} - \sin\theta\sigma_x^{\otimes 3},$$
$$A_3 = B_1 = \sigma_z^{\otimes 3}, \quad A_4 = B_2 = \sigma_x^{\otimes 3},$$
$$B_5 = \sigma_z^{\otimes 4}, \quad B_6 = \sigma_x^{\otimes 4},$$
$$C_1 = \sigma_z \otimes \sigma_z, \quad C_2 = \sigma_x \otimes \sigma_x. \tag{39}$$

From a forward evaluation we get

$$\varpi_q := \langle A_1B_1C_1\rangle + \langle A_2B_1C_1\rangle + \langle A_1B_2C_2\rangle - \langle A_2B_2C_2\rangle$$
$$+\langle A_3B_3C_1\rangle + \langle A_3B_4C_1\rangle + \langle A_4B_3C_2\rangle - \langle A_4B_4C_2\rangle$$

$$
\begin{aligned}
&+\langle A_3B_5C_3\rangle + \langle A_3B_5C_4\rangle + \langle A_4B_6C_3\rangle - \langle A_4B_6C_4\rangle \\
&= 6\cos\theta + 6\sin\theta\sin 2\theta_1\sin 2\theta_2\sin 2\theta_3 \\
&> 4\sqrt{2}+2, \qquad (40)
\end{aligned}
$$

which violates inequality (19) when θ_i's satisfy

$$\sin 2\theta_1 \sin 2\theta_2 \sin 2\theta_3 > 0.7928. \qquad (41)$$

Moreover, consider the noisy network $\mathcal{N}_q$ given by

$$\rho = \rho_1 \otimes \rho_2 \otimes \rho_3, \qquad (42)$$

where ρ_i denotes a Werner state [40] given by $\rho_i = v|\phi_i\rangle\langle\phi_i| + \frac{1-v}{4}\mathbb{1}$ with $\theta_i = \frac{\pi}{4}$, and $v \in (0, 1]$. The visibility is given by $v = 0.966$. This is going beyond a recent result without the robustness [32].

Continuing Example 2. Assume that two parties of Alice, Bob, and Charlie share a generalized EPR state: $|\phi_i\rangle_{2i-1,2i} = \cos\theta_i|00\rangle + \sin\theta_i|11\rangle$ with $\theta_i \in (0, \pi)$ and $i = 1$, 2, and 3. The total system is denoted as $|\phi_1\rangle_{12}|\phi_2\rangle_{34}|\phi_3\rangle_{56}$. Alice owns particles 1 and 6. Bob owns particles 2 and 3 while Charlie owns particles 4 and 5. To verify the genuinely tripartite nonlocality by using inequality (20), Alice and Bob will change $|\phi_1\rangle_{12}$ into a four-particle GHZ state $|\phi_1\rangle_{12ab} = \cos\theta_1|0000\rangle + \sin\theta_1|1111\rangle$ with local two-particle unitary operations for inputs 3 and 4. Define observables of three observers as follows:

$$
\begin{aligned}
A_1 &= \cos\theta\sigma_z\otimes\sigma_z + \sin\theta\sigma_x\otimes\sigma_z, \\
A_2 &= \cos\theta\sigma_z\otimes\sigma_z - \sin\theta\sigma_x\otimes\sigma_z, \\
A_3 &= \sigma_z^{\otimes 3}, \quad A_4 = \sigma_x^{\otimes 3}, \\
B_1 &= \sigma_z\otimes\sigma_z, \quad B_2 = \sigma_x\otimes\sigma_x, \\
B_3 &= \cos\theta\sigma_z^{\otimes 3} + \sin\theta\sigma_x^{\otimes 3}, \\
B_4 &= \cos\theta\sigma_z^{\otimes 3} - \sin\theta\sigma_x^{\otimes 3}, \\
C_1 &= \sigma_z\otimes\sigma_z, \quad C_2 = \sigma_x\otimes\sigma_z, \\
C_3 &= \cos\theta\sigma_z\otimes\sigma_z + \sin\theta\sigma_x\otimes\sigma_z, \\
C_4 &= \cos\theta\sigma_z\otimes\sigma_z - \sin\theta\sigma_x\otimes\sigma_z. \qquad (43)
\end{aligned}
$$

From a forward evaluation we get

$$
\begin{aligned}
\varpi_q :=\ & \langle A_1B_1C_1\rangle + \langle A_2B_1C_1\rangle + \langle A_1B_2C_2\rangle \\
& -\langle A_2B_2C_2\rangle + \langle A_3B_3C_1\rangle + \langle A_3B_4C_1\rangle \\
& +\langle A_4B_3C_2\rangle - \langle A_4B_4C_2\rangle + \langle A_3B_1C_3\rangle \\
& +\langle A_3B_1C_4\rangle + \langle A_4B_2C_3\rangle - \langle A_4B_2C_4\rangle \\
=\ & 6\cos\theta + 2\sin\theta(\sin 2\theta_1\sin 2\theta_2 \\
& + \sin 2\theta_2\sin 2\theta_3 + \sin 2\theta_1\sin 2\theta_3) \\
>\ & 4\sqrt{2}+2, \qquad (44)
\end{aligned}
$$

which violates inequality (20), where θ_i's satisfy

$$
\begin{aligned}
&\sin 2\theta_1\sin 2\theta_2 + \sin 2\theta_2\sin 2\theta_3 + \sin 2\theta_1\sin 2\theta_3 \\
&> 2\times 2^{1/4} \\
&\approx 2.3784. \qquad (45)
\end{aligned}
$$

For the special case of $\theta_1 = \theta_2 = \theta_3$, it follows that $\theta > \frac{1}{2}\arcsin((\frac{32}{81})^{1/8}) \approx 0.1748\pi$, where θ satisfies $\cos\theta = \frac{1}{\sqrt{1+(\sin 2\theta)^4}} \approx 0.59$. If $\theta_1 = \theta_2 = \frac{\pi}{4}$, it follows that $\theta_3 > 0.121\pi$. The same visibility can be followed from inequalities (19) and (20).

E. Robustness of genuinely multipartite nonlocality

The goal in this section is to prove the genuinely multipartite nonlocality for noisy networks. Consider an n-partite connected network $\mathcal{N}_q$ consisting of noisy states ρ_i, ϱ_j, where ρ_i and ϱ_i are Werner states [40] given respectively by

$$
\begin{aligned}
\rho_i &= v_i|\phi_i\rangle\langle\phi_i| + \frac{1-v_i}{4}\mathbb{1}_4, \\
\varrho_i &= w_i|\psi_i\rangle\langle\psi_i| + \frac{1-w_i}{2^{k_i}}\mathbb{1}_{2^{k_i}}, \qquad (46)
\end{aligned}
$$

where $|\phi_i\rangle$ is an EPR state, $|\psi_j\rangle$ is a GHZ state with k_j particles, $\mathbb{1}_j$ is the identity operator with the rank j, $v_i, w_j \in (0, 1]$, $i = 1, \ldots, m_1; j = 1, \ldots, m_2$.

Theorem 4. The network $\mathcal{N}_q$ is genuinely n-partite nonlocal if v_i, w_j satisfy the following inequality:

$$\prod_{i=1}^{m_1}\prod_{j=1}^{m_2} v_i w_j > 1 - \frac{2-\sqrt{2}}{2N} \qquad (47)$$

with $N = 2^{n-1} - 1$.

Proof. The proof is similar to those stated in Secs. II B and II C. Note that all observables are dichotomic. We get $CH_I' = \prod_{i=1}^{m_1}\prod_{j=1}^{m_2} v_i w_j CH_I$, where CH_I is defined in inequality (2) for the network $\mathcal{N}_q$ without noise while $CH_{\vec{s}_i}'$ denotes the same quantity associated with a noisy network shown in Eq. (46). From Eqs. (22)–(32) it implies that

$$
\begin{aligned}
\varpi_q &= \sum_i \mathcal{L}_i \\
&\geqslant 2N\sqrt{2}\prod_{i=1}^{m_1}\prod_{j=1}^{m_2} v_i w_j \\
&> 2^n\sqrt{2} - 4\sqrt{2} + 2 \qquad (48)
\end{aligned}
$$

when v_i, w_j satisfy the inequality of $\prod_{i=1}^{m_1}\prod_{j=1}^{m_2} v_i w_j > 1 - \frac{2-\sqrt{2}}{2N}$. Numeric simulations are shown in Fig. 3 for the visibility of n-partite cyclic networks. This completes the proof. ■

III. A DEVICE-INDEPENDENT MODEL ON QUANTUM NETWORKS

For single entangled states, the violation of a certain Bell inequality allows for device-independent information processing against quantum (postquantum) eavesdroppers [5–11,28–30,33,42,43]. Our goal in this section is to propose a device-independent model using quantum networks based on inequality (2).

Consider a connected network $\mathcal{N}$ [Fig. 2(a)]. For a device-independent model, an eavesdropper may hold local systems correlated with some sources in order to recover private information such as outcomes.

Denote the variation distance of two probability distributions $\{p(x)\}$ and $\{q(x)\}$ as $D(p, q) = \frac{1}{2}\sum_x |p(x) - q(x)|$.

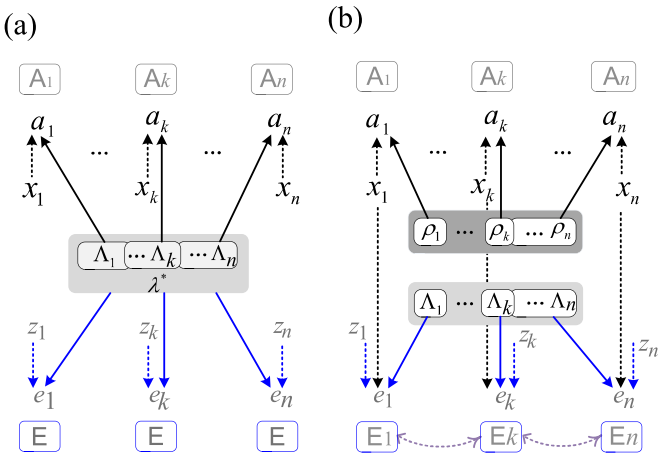


FIG. 2. (a) A device-independent model on an n-partite network $\mathcal{N}$ consisting of m sources $\lambda_1, \ldots, \lambda_m$. A quantum (postquantum) eavesdropper E holds some systems correlated with the shared sources. z_i and e_i denote the respective input and outcome of the eavesdropper for recovering the output a_i. For a device-independent model, E may correlate λ_i's into a new source λ^*. (b) Consider a quantum network $\mathcal{N}_q$, where A_i shares a system ρ_i consisting of some EPR states and GHZ states with A_j, $i = 1, \ldots, k$, and $j = k+1, \ldots, n$. E_i's are eavesdroppers sharing random variables Λ_i consisting of λ_j with the same network configuration as $\mathcal{N}_q$. e_i is the output of E_i conditional on the inputs z_i and x_i and the variables Λ_i. E_i's want to classically simulate quantum correlations derived from local measurements on $\mathcal{N}_q$.

Theorem 5

The predictive power of an eavesdropper to learn the outcomes of legitimate parties satisfies

$$D_e \leqslant 2n - \frac{\varpi_q - 2\sqrt{2}(\alpha - \lfloor\alpha\rfloor)}{2\lfloor\alpha\rfloor}, \tag{49}$$

where $D_e = D(\prod_{i=1}^n P(e_i|a_i;\mathbf{x},z_i), \prod_{i=1}^n P(e_i|z_i))$, $\alpha = \frac{1}{n}(2^n-1)$, $\mathbf{x} = x_1\cdots x_n$, $\varpi_q = \sum_{i=1}^{\lfloor\frac{n}{2}\rfloor}\mathcal{L}_i$, and $\mathcal{L}_i$ is defined in inequality (2) with the estimated quantum correlations.

Proof. In this section, we prove inequality (49) for connected quantum networks. Consider a general network $\mathcal{N}$ (Fig. 2) consisting of n parties $\mathsf{A}_1, \ldots, \mathsf{A}_n$, who share m sources $\lambda_1, \ldots, \lambda_m$. An eavesdropper may gain information related to the outcome of A_i by measuring the correlated system with input z_i and outcome e_i. Suppose that all the inputs of $\mathbf{x} = x_1\cdots x_n$ are accessible to the eavesdropper. Note that in secure applications all the legal parties will not send out their measurement outcomes before they begin to detect the eavesdropper. This means that a smart eavesdropper will recover one outcome a_i by using the prior knowledge of $a_i, \mathbf{x}, z_i$, i.e., $P(e_i|a_i,\mathbf{x},z_i)$. Hence, we consider the variation distance of two probability distributions $\{\prod_i P(e_i|a_i,\mathbf{x},z_i)\}$ and $\{\prod_i P(e_i|\mathbf{x},z_i)\}$. From the nonsignaling condition we get that

$$\begin{aligned} P(e_i|\mathbf{x},z_i) &= P(e_i|z_i), \\ P(e_i|a_i,\mathbf{x},z_i) &= P(e_i|a_i,x_i,z_i). \end{aligned} \tag{50}$$

Denote $D_e := D(\prod_{i=1}^n P(e_i|a_i;\mathbf{x},z_i), \prod_{j=1}^n p(e_j|z_j))$ for convenience. From Eq. (50), the left-hand side of inequality (49) can be rewritten into

$$\begin{aligned} D_e \leqslant{}& \frac{1}{2}\sum_{\mathbf{e}} p(e_1|a_1;\mathbf{x},z_1)\left|\prod_{i=2}^n p(e_i|a_i;\mathbf{x},z_i) - \prod_{i=2}^n p(e_i|z_i)\right| \\ &+\frac{1}{2}\sum_{\mathbf{e}} |p(e_1|a_1;\mathbf{x},z_1) - p(e_1|z_1)|\prod_{i=2}^n p(e_i|z_i). \end{aligned} \tag{51}$$

Here, inequality (51) is followed from the triangle inequality $|x-y| \leqslant |x-z| + |z-y|$. Moreover, we have

$$\begin{aligned} D_e &\leqslant D(p(e_1|a_1;\mathbf{x},z_1), p(e_1|z_1)) \\ &\quad + D\left(\prod_{i=2}^n p(e_i|a_i;\mathbf{x},z_i), \prod_{i=2}^n p(e_i|z_i)\right) && (52) \\ &\leqslant \sum_{i=1}^n D(p(e_i|a_i;\mathbf{x},z_i), p(e_i|z_i)) && (53) \\ &= \sum_{i=1}^n D(p(e_i|a_i;x_i,x_j,z_i), p(e_i|z_i)) && (54) \\ &\leqslant \sum_{i=1}^n I_2(P(a_i,a_j|x_i,x_j). && (55) \end{aligned}$$

Inequality (52) is from the normalization conditions of $\sum_{e_1} p(e_1|z_1) = 1$, and $\sum_{e_j} p(e_j|a_j;\mathbf{x},z_j) = 1$, $j = 2, \ldots, n$. Inequality (53) is obtained by a similar process of inequality (51) with $n-1$ iterations. Equation (54) follows from Eq. (50), where A_i and A_j share at least one hidden variable. Inequality (55) is obtained by using the inequality of $D(p(e_i|a_i;x_i,x_j,z_i), p(e_i|z_i)) \leqslant I_2(P(a_i,a_j|x_i,x_j))$ with the chained Bell inequality [28,44,45], where $I_2(P(a_i,a_j|x_i,x_j)) = P(a_i = a_j|x_i = 0; x_j = 1) + \sum_{|x_i-x_j|=1} P(a_i \neq a_j|x_i;x_j)$, x_i and a_i are the input and outcome of A_i, and x_j and a_j are the input and outcome of A_j.

Consider a quantum network $\mathcal{N}_q$ on which all parties have multiple inputs and binary outcomes. The quantum violation ϖ_{es} of practical quantum correlations is shown as follows:

$$\begin{aligned} \varpi_{es} &= \sum_{i=1}^{\lfloor\frac{n}{2}\rfloor}\sum_I [\langle(\mathbf{M}_I + \hat{\mathbf{M}}_I)\mathbf{M}_{\bar{I}}\rangle + \langle(\mathbf{M}_I - \hat{\mathbf{M}}_I)\hat{\mathbf{M}}_{\bar{I}}\rangle] \\ &\leqslant \sum_{i=1}^{\lfloor\frac{n}{2}\rfloor}\sum_I [|\langle(\mathbf{M}_I + \hat{\mathbf{M}}_I)\mathbf{M}_{\bar{I}}\rangle| + |\langle(\mathbf{M}_I - \hat{\mathbf{M}}_I)\hat{\mathbf{M}}_{\bar{I}}\rangle|] \\ &\leqslant \sum_{i=1}^{\lfloor\frac{n}{2}\rfloor}\sum_{j\in I} \big[|\langle(M_{x_j(I)} + \hat{M}_{x_j(I)})M_{x_{k_j}(\bar{I})}\rangle| \\ &\quad + \big|\langle(M_{x_j(I)} - \hat{M}_{x_j(I)})\hat{M}_{x_{k_j}(\bar{I})}\big|\big] && (56) \\ &= \sum_{i=1}^{\lfloor\frac{n}{2}\rfloor}\sum_I CH_{M_jM_{k_j}}. && (57) \end{aligned}$$

Here, inequality (56) follows from the inequality of $\langle A\rangle \leqslant |\langle A\rangle|$. Inequality (56) is from the assumptions of separable observables ($M_{x_i} \leqslant 1$) except for $M_{x_{k_j}(\bar{I})}$ and $\hat{M}_{x_{k_j}(\bar{I})}$ given in Eq. (27). $M_{x_{k_j}}$ and $\hat{M}_{x_{k_j}}$ denote two observables of $\mathsf{A}_{x_{k_j}}$ with

$k_j \notin I$ associated with $j \in I$. $CH_{M_jM_{k_j}}$ in Eq. (57) is defined by

$$CH_{M_jM_{k_j}} = \langle (M_{x_j(I)} + \hat{M}_{x_j(I)}) M_{x_{k_j}(\bar{I})} \rangle + \langle (M_{x_j(I)} - \hat{M}_{x_j(I)}) \hat{M}_{x_{k_j}(\bar{I})} \rangle,$$

where we have $\langle (M_{x_j(I)} + \hat{M}_{x_j(I)}) M_{x_{k_j}(\bar{I})} \rangle \geqslant 0$ and $\langle (M_{x_j(I)} - \hat{M}_{x_j(I)}) \hat{M}_{x_{k_j}(\bar{I})} \rangle \geqslant 0$ in applications.

From the evaluations in Eqs. (32) and (34), we obtain that

$$\varpi_{es} \leqslant \left\lfloor \frac{N}{n} \right\rfloor \sum_{j=1}^{n} CH_{M_jM_{k_j}} + 2\sqrt{2}\left(\frac{N}{n} - \left\lfloor \frac{N}{n} \right\rfloor \right), \quad (58)$$

where $N = 2^n - 1$. From $\langle XY \rangle = 2p(X = Y) - 1$ it follows that $I_2(P(a_i, a_j|x_i, x_j)) = 2 - \frac{1}{2} CH_{X_iX_j}$. Using inequalities (55) and (58), D_e is evaluated as follows:

$$D_e \leqslant 2n - \frac{1}{2} \sum_{j=1}^{n} CH_{M_jM_{k_j}} \leqslant 2n - \frac{\varpi_{es} - 2\sqrt{2}(n_0 - \lfloor n_0 \rfloor)}{2\lfloor n_0 \rfloor} \quad (59)$$

with $n_0 = \frac{N}{n}$. This completes the proof. ■

Inequality (49) provides a monogamy relation for the information leakage in a device-independent model on quantum networks.

IV. CLASSICAL SIMULATION OF QUANTUM CORRELATIONS WITH FINITE CLASSICAL COMMUNICATION AND SHARED RANDOMNESS

There are two assumptions for guaranteeing the security against quantum or postquantum eavesdroppers. One is that an eavesdropper may access all inputs, and the other is that an eavesdropper (not a party) cannot communicate with other parties. Otherwise, the bipartite quantum correlations can be precisely simulated with shared classical random variables [46,47]. A similar result holds for multisource quantum networks. In fact, suppose that a general network $\mathcal{N}_q$ in Fig. 2(b) is k independent; that is, there are k parties $\mathsf{A}_1, \ldots, \mathsf{A}_k$ (for simplicity) who do not share any entangled state with each other. $\mathcal{N}_q$ is schematically represented by a DAG [Fig. 1(b)] with the same configuration as $\mathcal{N}_q$ using random variables $\lambda_1, \ldots, \lambda_m$. $(\Omega_i, \Sigma_i, \mu_i)$ denotes the measure space of λ_i with $i = 1, \ldots, m$. Each observer A_i obtains states from some random sources $\Lambda_i = \{\lambda_{j_1}, \ldots, \lambda_{j_{t_i}}\}$. The outcome a_i of A_i depends on the shared variables Λ_i and the type of local measurement x_i. The joint conditional distributions of the outcomes conditional on the inputs are given as

$$P_c(\mathbf{a}|\mathbf{x}) = \int_{\times_{i=1}^{m}\Omega_i} \prod_{j=1}^{n} P_c(a_j|x_j, \Lambda_j) \prod_{i=1}^{m} d\mu_i(\lambda_i). \quad (60)$$

Define a classical correlation as

$$E_c(\vec{\mathbf{x}}) = \sum_{\Xi_1} P_c(\mathbf{a}_k|\vec{\mathbf{x}}_k) - \sum_{\Xi_2} P_c(\mathbf{a}_k|\vec{\mathbf{x}}_k), \quad (61)$$

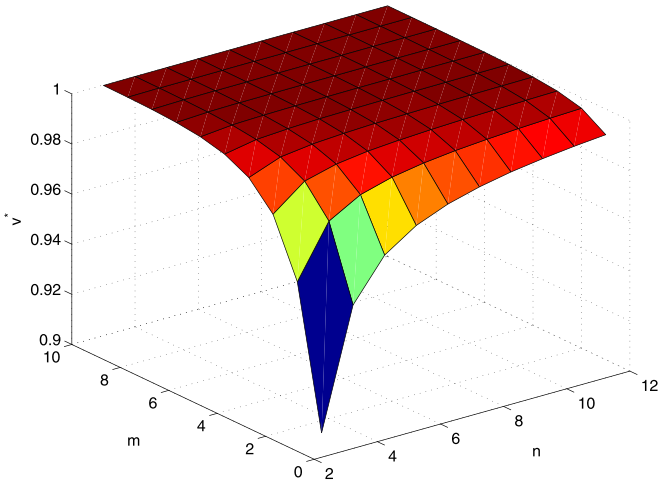


FIG. 3. Visibility of Werner states. Here, all v_i's and w_j's are equal to each other. n denotes the number of parties while $m = m_1 + m_2$ denotes the number of entangled states.

where $P_c(\mathbf{a}_k|\vec{\mathbf{x}}_k)$ denotes the marginal probability of $\mathbf{a}_k$ conditional on $\vec{\mathbf{x}}_k$, and Ξ_1 denotes all combinations with even number of output -1 while Ξ_2 denotes all combinations with odd number of output -1, $\mathbf{a} = a_1 \cdots a_n$, $\mathbf{x} = x_1 \cdots x_n$, $\mathbf{a}_k = a_1 \cdots a_k$, and $\vec{\mathbf{x}}_k = \vec{x}_1 \cdots \vec{x}_k$.

Consider special applications with a joint system in the state ρ, such as multipartite QKD. All the independent parties of $\mathsf{A}_1, \ldots, \mathsf{A}_k$ will finally perform qubit measurements after the others' local measurements and classical communication. The joint conditional probability $P(\mathbf{a}|\vec{\mathbf{x}}_k, x_{k+1} \cdots x_n)$ is defined as

$$P_q(\mathbf{a}|\vec{\mathbf{x}}_k, x_{k+1} \cdots x_n) = \mathrm{tr}(\vec{x}_1 \otimes \cdots \otimes \vec{x}_n \rho), \quad (62)$$

where $\vec{x}_i$ is a qubit measurement vector under the Pauli basis $(\sigma_x, \sigma_y, \sigma_z)$. Similar to Eq. (61), define a quantum correlation as

$$E_q(\vec{\mathbf{x}}) = \sum_{\Xi_1} P_q(\mathbf{a}_k|\vec{\mathbf{x}}_k) - \sum_{\Xi_2} P_q(\mathbf{a}_k|\vec{\mathbf{x}}_k). \quad (63)$$

Consider a protocol completed in a device-independent manner as shown in Fig. 3(b). Suppose that E_i's share variables Λ_i consisting of independent random variables λ_j with the same network configuration as $\mathcal{N}_q$; that is, all the parties A_i are replaced with E_i in the hidden variable model of $\mathcal{N}_q$. The goal in this section is to simulate the quantum correlation $E_q(\vec{\mathbf{x}})$ in Eq. (63) by using the classical correlation $E_c(\vec{\mathbf{x}})$ in Eq. (61). If all inputs are achievable for eavesdroppers as shown in Fig. 2(b), the quantum correlations derived from $\mathcal{N}_q$ may be classically simulated with shared randomness and finite classical communication. This partially answers a recent conjecture [47]. It inspires special restraints on the measurements to enhance the security for cryptographic applications.

Theorem 6. Suppose that eavesdroppers E_i can access all measurement inputs $\vec{x}$. E_i's can generate a classical correlation $E_c(\vec{\mathbf{x}})$ such that

$$E_c(\vec{\mathbf{x}}) = E_q(\vec{\mathbf{x}}) \quad (64)$$

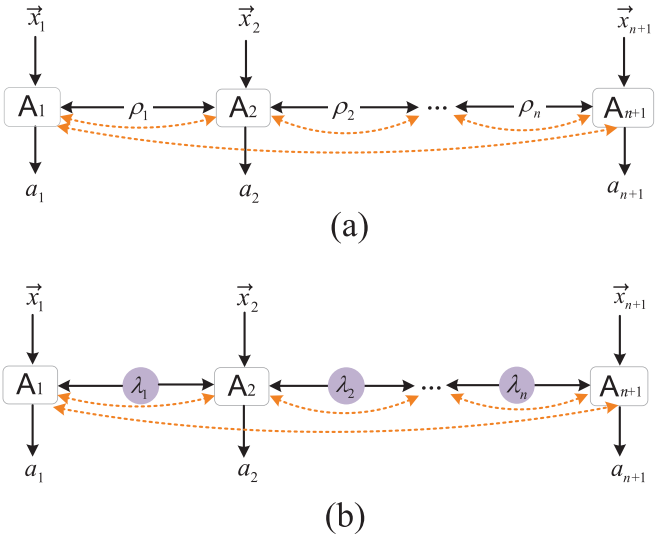


FIG. 4. Schematic of a long-distance entanglement swapping network. (a) Long-distance entanglement swapping network consisting of k independent sources $\rho_i = |\phi_i\rangle\langle\phi_i|$ with generalized EPR states $|\phi_i\rangle = \alpha_i|01\rangle - \beta_i|10\rangle$. All the middle observers $\mathsf{A}_2, \ldots, \mathsf{A}_k$ can help A_1 and A_{k+1} to build a single entangled state with local operations and classical communication (LOCC). (b) Classical simulation of correlations generated by a long-distance entanglement swapping experiment. Eavesdroppers share n random variables $\lambda_1, \ldots, \lambda_k$. In a simulation experiment, eavesdroppers can access all measurement inputs and finite classical communication. The red dashed arrows represent exchange of classical messages.

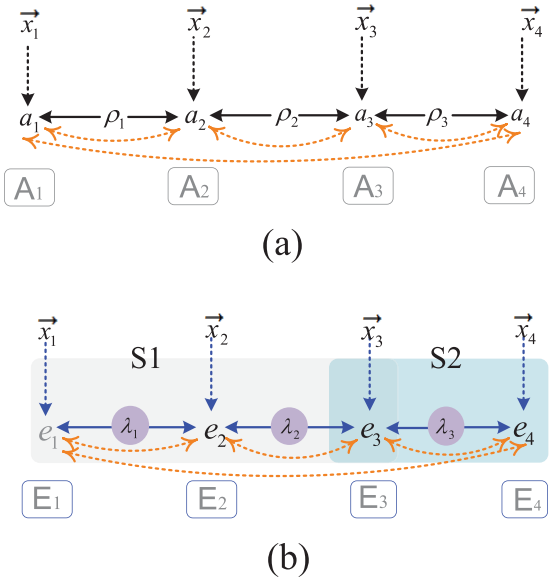


FIG. 5. Classical simulation of quantum correlations from entanglement swapping. (a) Entanglement swapping with four observers $\mathsf{A}_1, \ldots, \mathsf{A}_4$, who share EPR states $\rho_1 \otimes \rho_2 \otimes \rho_3$. (b) The classical simulation of quantum correlations generated by entanglement swapping by eavesdroppers $\mathsf{E}_1, \ldots, \mathsf{E}_4$ who share random variables $\lambda_1, \lambda_2, \lambda_3$. The classical communication is represented by red dashed arrows. The classical simulation is completed by two simulations S1 and S2 without assumptions of correlating sources.

by using shared randomness and finite classical communication, where the classical correlations are defined by $E_c(\vec{\mathbf{x}}) = \sum_{\Xi_1} P_c(\mathbf{e}_k|\vec{\mathbf{x}}_k) - \sum_{\Xi_2} P_c(\mathbf{e}_k|\vec{\mathbf{x}}_k)$, $P_c(\mathbf{e}_k|\vec{\mathbf{x}}_k)$ denotes the joint distribution of eavesdroppers $\mathsf{E}_1, \ldots, \mathsf{E}_k$ conditional on the inputs $\vec{\mathbf{x}}_k$, and their own inputs $\mathbf{e}_k = e_1 \cdots e_k$.

Proof. Consider a general quantum network $\mathcal{N}_q$ [Fig. 2(a)], which consists of all entangled pure states. Suppose that there are k independent observers $\mathsf{A}_1, \ldots, \mathsf{A}_k$, who do not share entangled states. Our goal here is to show a classical simulation of the multipartite correlations by eavesdroppers who can access all inputs and commutate with each other. A classical simulation with the same network configuration is shown in Fig. 2(b). One example of Fig. 2 is the long-distance entanglement swapping network given in Fig. 4(a). In experiment, all the middle observers $\mathsf{A}_2, \ldots, \mathsf{A}_k$ perform Bell measurements $\sigma_z \otimes \sigma_z$, $\sigma_x \otimes \sigma_x$ and send the outcomes to A_1 and A_{k+1} who can recover a single entangled state with local unitary operations. To complete the simulation shown in Fig. 4(b), suppose that A_1 and A_{k+1} perform the qubit projective measurements represented by the respective Bloch vector $\vec{x}_1, \vec{x}_{k+1}$ in terms of the Pauli basis $(\sigma_x, \sigma_y, \sigma_z)$. Suppose that A_1 and A_{k+1} obtain binary outcomes $a_1, a_{k+1} = \pm 1$, respectively. The correlations of A_1 and A_{k+1} exhibit the following form:

$$E_q(\vec{x}_1, \vec{x}_{k+1}) := \sum_{\Xi_1} P_q(\mathbf{a}|\vec{\mathbf{x}}) - \sum_{\Xi_2} P_q(\mathbf{a}|\vec{\mathbf{x}}), \tag{65}$$

where $P_q(\mathbf{a}|\vec{\mathbf{x}})$ is defined in Eq. (63). An open problem [3] is to determine the classical simulation capability of $E_q(\vec{x}_1, \vec{x}_{k+1})$ for $n \geqslant 3$ with independent sources and finite communication.

Generally, the proof of Eq. (64) is divided into three cases. One is to prove the classical simulation of long-distance entanglement swapping shown in Fig. 4(a) with EPR states for $k \geqslant 1$. The second is to prove the classical simulation of generalized entanglement swapping for any acyclic networks consisting of EPR states shown in Fig. 5. The last one is to prove the result for any acyclic networks consisting of generalized EPR states shown in Fig. 6. In what follows, we only present the proof for the first case while the others are shown in Appendix B. For $k = 1$, the network shown in Fig. 4(a) reduces to a standard Bell network [46]. For $k = 2$, it is a standard entanglement swapping network that can be simulated classically [47]. In what follows, we prove the result for $k > 2$. The proof is completed by induction k.

We first prove the result for $k = 3$, as shown in Fig. 5. A_i and A_{i+1} share one or two EPR states shown in Fig. 5(a). A_1 and A_4 obtain the respective measurement input $\vec{x}_1 = (\cos\theta_a, 0, \sin\theta_a)$ and $\vec{x}_4 = (\cos\theta_b, 0, \sin\theta_b)$. The correlation given in Eq. (65) is rewritten into

$$\begin{aligned} E_q(\vec{x}_1, \vec{x}_4) &= \sum_{\Xi_1} P_q(\mathbf{a}|\vec{\mathbf{x}}) - \sum_{\Xi_2} P_q(\mathbf{a}|\vec{\mathbf{x}}) \\ &= \cos(\theta_a - \theta_b). \end{aligned} \tag{66}$$

The classical simulation is shown in Fig. 5(b), where E_i and E_{i+1} share one or two random variables λ_i with the same network configuration as shown in Fig. 5(a). The finite classi-

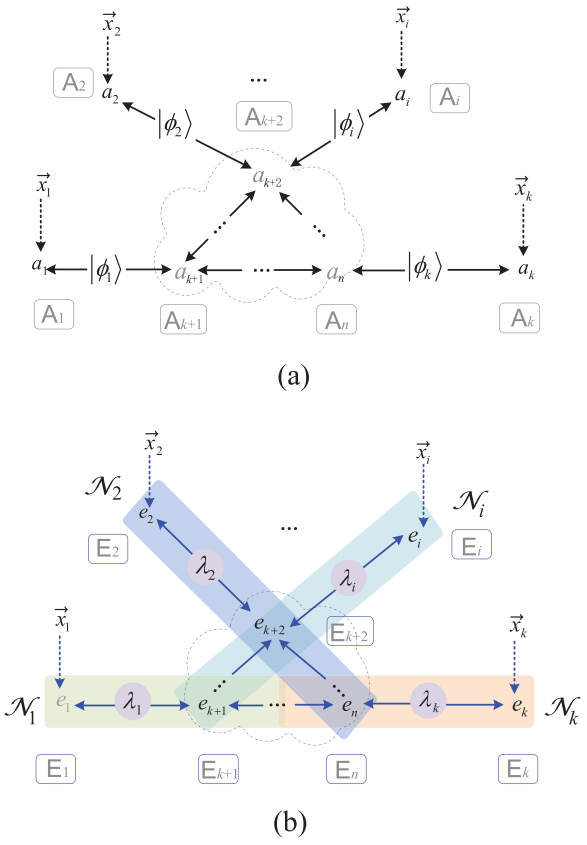


FIG. 6. Classical simulation of quantum correlations from generalized entanglement swapping. (a) A generalized entanglement swapping on any acyclic connected quantum network $\mathcal{N}_q$. The resources consist of generalized EPR states $\otimes_i|\phi_i\rangle$ with $|\phi_i\rangle = \alpha_i|01\rangle + \beta_i|10\rangle$. $\mathsf{A}_{k+1},\ldots,\mathsf{A}_n$ perform the multiple-particle Bell measurement on local systems and send outcomes to other observers $\mathsf{A}_1,\ldots,\mathsf{A}_k$, who can recover a multipartite generalized GHZ state by proper local unitary operations. (b) The classical simulation of generalized entanglement swapping by nonsignaling eavesdroppers $\mathsf{E}_1,\ldots,\mathsf{E}_n$ who share variables λ_i. The red dashed arrows represent exchanging classical messages. The classical simulation network is decomposed into k chain-shaped subnetworks $\mathcal{N}_1,\ldots,\mathcal{N}_k$. Finite classical communication is allowed for all eavesdroppers.

cal communication is allowed for E_i's who know the inputs of A_1 and A_4. The simulation protocol consists of two separable simulations S1 and S2 as follows:

(S1) E_i's set up the first classical simulation according to finite classical communication.

(a) E_1, E_2, and E_3 obtain a classical correlation as follows:

$$E_c(\vec{x}_1',\vec{y}_1) = \sum_{\Xi_1} P_c(\mathbf{e}|\vec{x}_1',\vec{y}_1) - \sum_{\Xi_2} P_c(\mathbf{e}|\vec{x}_1',\vec{y}_1) = 2\cos\theta_a \tag{67}$$

from a tripartite classical simulation protocol [47] with independent sources and finite communication. Here, the inputs are assumed to be $\vec{x}_1' = (2\cos\theta_a, 0, 0)$ and $\vec{y}_1 = (1, 0, 0)$.

(b) E_3 and E_4 obtain a classical correlation as follows:

$$E_c(\vec{y}_1,\vec{x}_4') = P_c(e_3 = e_4|\vec{y}_1,\vec{x}_4') - P_c(e_3 \neq e_4|\vec{y}_1,\vec{x}_4') = \cos\theta_b \tag{68}$$

from a bipartite classical simulation protocol [46], where $\vec{x}_4' = (\cos\theta_b, 0, 0)$. Note that two steps are independent. It follows that

$$E_c(\vec{x}_1',\vec{x}_4') = E_c(\vec{x}_1',\vec{y}_1)E_c(\vec{y}_1,\vec{x}_4') = 2\cos\theta_a\cos\theta_b, \tag{69}$$

which can be easily followed from the separable Bell measurement of A_3, i.e., $\langle M_{x_1}M_{x_2}M_{x_3}M_{x_4}\rangle = \langle M_{x_1}M_{x_2}M_{x_3;1}\rangle\langle M_{x_3;2}M_{x_4}\rangle$ with separable measurement $M_{x_3;1}\otimes M_{x_3;2}$ of A_3, where M_{x_1} is observable with dichotomic outcomes. Equivalently, one can prove the result from the conditional independence of full correlations $P_q(\mathbf{e}|\vec{x}) = P_q(\mathbf{e}_1|\vec{x}_1)P_q(\mathbf{e}_2|\vec{x}_2)$, where $P_q(\mathbf{e}_1|\vec{x}_1)$ and $P_q(\mathbf{e}_2|\vec{x}_2)$ denote the respective correlation obtained from the tripartite network (consisting of E_1, E_2, and E_3) and the bipartite network (consisting of E_3 and E_4). Here, E_3 in Fig. 5(b) or A_3 in Fig. 5(a) performs the separable measurement.

(S2) E_i's set up the second classical simulation according to finite classical communication.

(a) E_1, E_2, and E_3 obtain a classical correlation as follows:

$$E_c(\vec{x}_1'',\vec{y}_1) = \sum_{\Xi_1} P_c(\mathbf{e}|\vec{x}_1'',\vec{y}_1) - \sum_{\Xi_2} P_c(\mathbf{e}|\vec{x}_1'',\vec{y}_1) = 2\sin\theta_a \tag{70}$$

from a tripartite classical simulation protocol [47] by using independent sources and finite communication, where their inputs are assumed to be $\vec{x}_1'' = (0, 0, 2\sin\theta_a)$ and $\vec{y}_1 = (0, 0, 1)$.

(b) E_3 and E_4 obtain a classical correlation as follows:

$$E_c(\vec{y}_1,\vec{x}_4'') = P_c(e_3 = e_4|\vec{y}_1,\vec{x}_4'') - P_c(e_3 \neq e_4|\vec{y}_1,\vec{x}_4'') = \sin\theta_b \tag{71}$$

from a bipartite classical simulation protocol [46] without correlating sources, where $\vec{x}_4'' = (0, 0, \sin\theta_b)$. These two steps are independent. It follows that

$$E_c(\vec{x}_1'',\vec{x}_4'') = E_c(\vec{x}_1'',\vec{y}_1)E_c(\vec{y}_1,\vec{x}_4'') = 2\sin\theta_a\sin\theta_b. \tag{72}$$

From Eqs. (69) and (72), it follows that

$$\sum_{\Xi_1} P_c(\mathbf{e}|\vec{\mathbf{x}}) - \sum_{\Xi_2} P_c(\mathbf{e}|\vec{\mathbf{x}}) := \frac{1}{2}(E_c(\vec{x}_1',\vec{x}_4') + E_c(\vec{x}_1'',\vec{x}_4'')) = \cos(\theta_a-\theta_b) = E_q(\vec{x}_1,\vec{x}_4) \tag{73}$$

from Eq. (66), where S1 and S2 are two independent simulations which are recombined into one simulation with equal probability. No correlating operation is required for any sources. In this protocol, total classical communication is finite because each simulation uses finite communication.

Now, by induction we can prove the result for any $k \geqslant 3$, where the classical simulations are divided into two separable chained subnetworks consisting of $\mathsf{E}_1,\ldots,\mathsf{E}_k$, and E_k and E_{k+1}, respectively. A similar result holds for generalized measurement in terms of the Pauli basis [46,47] when all measurement inputs are accessible to eavesdroppers

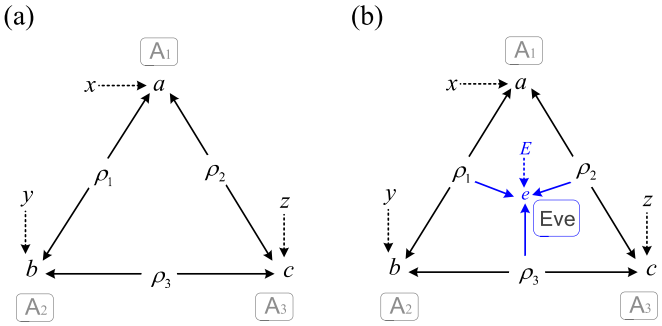


FIG. 7. Device-independent tripartite QKD. (a) Tripartite QKD based on a triangle network. a, b, and c (x, y, and z) denote the output (input) of A_1, A_2, and A_3, respectively. (b) Device-independent tripartite QKD. The output e of eavesdropper E depends on its input E and these shared systems.

without assumption of correlating sources. This completes the proof. ■

Remark 2. The classical simulation requires only finite classical communication for E_i's [see Fig. 2(b)]. Here, eavesdroppers cannot simulate the joint distribution $P(\mathbf{a}|\mathbf{x})$ or the exact outputs $\mathbf{a}$ for each set of inputs $\mathbf{x}$. That is, the present simulation in Eq. (64) provides only potential information leakage of correlations based on generalized entanglement swapping going beyond chain-shaped networks without correlating assumptions [47]. However, it does not imply the information leakage of measurement outputs in device-independent applications.

V. DEVICE-INDEPENDENT MULTIPARTITE QUANTUM KEY DISTRIBUTION

Quantum entanglement is useful for distributing random keys for secure communication [5–8]. Our goal is to present a device-independent QKD with multiple parties using quantum networks. For simplicity, consider a tripartite network consisting of A_1, A_2, and A_3 [Fig. 7(a)]. Let x, y, and z, respectively, denote the inputs of the three parties, and a, b, and c denote the corresponding outcomes. ρ_{ABC} denotes the total system. From inequality (20), there are four inputs $x, y, z \in \{1, 2, 3, 4\}$ for each party while two outputs $a, b, c \in \{0, 1\}$ are for each input. In experiments, all the parties share N copies of a triangle network, that is, $\rho^{\otimes N}$. Each party A_1 (or A_2 or A_3), for example, performs a measurement A_{x_i} (or A_{y_i} or A_{z_i}) on its own particles according to its input x_i (or y_i or z_i) chosen uniformly at random by the user in the ith round. Their outputs are denoted as $a_i, b_i, c_i \in \{0, 1\}$. They choose a random subset $B \subseteq \{1, \ldots, N\}$ of test rounds of size γN and share it publicly with each other. All the parties announce their input-output pairs in all rounds in B. They compute the fraction of pairs in B that satisfy inequality condition (20). If this fraction is smaller than some threshold they abort the protocol. Otherwise, they publicly reveal their inputs. Denote C as the set of all the check rounds with $(x_i, y_i, z_i) \in \{(1, 1, 1), (2, 1, 1), (1, 2, 2), (2, 2, 2), (3, 3, 1), (3, 4, 1), (4, 3, 2), (4, 4, 2), (3, 1, 3), (3, 1, 4), (4, 2, 3), (4, 2, 4)\}$. They perform information reconciliation on the rounds in C, followed by privacy amplification, to obtain their respective key k_a, k_b, k_c from their outputs. These strings are further used to obtain the final shared keys by postprocessing error reconciliation and privacy amplification, but we do not discuss this standard procedure here. In application, one may consider some proper subset of inputs in order to further reduce information leakage. In a device-independent multipartite QKD (DIMQKD), the information available to an eavesdropper Eve is represented by E correlated with ρ_{ABC} [Fig. 7(b)]. The total state is then denoted by ρ_{ABCE} which should satisfy $\mathrm{tr}_E \rho_{ABCE} = \rho_{ABC}$. Here, suppose that Eve wants to learn the output a of A_1. For one copy resource, the predictability for Eve is quantified by the guessing probability of a as $P_{gs}(a) = \max_a P(a|x)$. The asymptotic secret-key rate is defined by $R := -\frac{\log_2 P_{gs}(\mathbf{a}|E)}{s}$ [29,48,49].

Theorem 7. The following results hold:

(i) The guessing probability is bounded by

$$P_{gs}(a) \leqslant \frac{1}{2} + \frac{1}{12}\sqrt{72 - \varpi_{es}^2}, \tag{74}$$

where ϖ_{es} denotes the estimate of inequality (2) using the practical quantum correlations in accordance with the practical quantum network.

(ii) The asymptotic secret-key rate satisfies

$$R \geqslant -\log_2\left(\frac{1}{2} + \frac{1}{12}\sqrt{72 - \varpi_{es}^2}\right) - H(a|b, c), \tag{75}$$

where $P_{gs}(\mathbf{a}|E)$ denotes the expected optimal guessing probability conditional on the eavesdropper's input E. The first term of Eq. (75) represents the knowledge of Eve for the privacy amplification, and $H(a|b, c)$ quantifies the information needed for the error correction by A_1.

Proof. From inequality (2), the expected violation of quantum correlations is represented by

$$\begin{aligned}\varpi_{es} = \sum_{\substack{a_i,b_i,c_i \\ x_i,y_i,z_i \in S_{in}}} & [(-1)^{a_1+b_1+c_1+x_1y_1z_1} P(a_1b_1c_1|x_1y_1z_1) \\ & +(-1)^{a_2+c_2+b_2+x_2y_2z_2} P(a_2b_2c_2|x_2y_2z_2) \\ & +(-1)^{a_3+b_3+c_3+x_3y_3z_3} P(a_3b_3c_3|x_3y_3z_3)],\end{aligned} \tag{76}$$

where $P(a_ib_ic_i|x_iy_iz_i) = \mathrm{tr}[(A_{x_i}^{a_i} B_{y_i}^{b_i} C_{z_i}^{c_i})\rho]$, $A_{x_i}^{a_i}$, $B_{y_i}^{b_i}$, and $C_{z_i}^{c_i}$ denote the POVM operators of A_1, A_2, and A_3, respectively, ρ denotes the total state of the triangle network (Fig. 3), and S_{in} denotes the input set satisfying the conditions $x_1 = y_1$, $x_2 = z_2$, and $y_3 = z_3$. From the symmetry of inputs x_i, y_i, and z_i, it follows that

$$\varpi_{es} = 3\sum_{\substack{a_1,b_1,c_1 \\ x_1,y_1,z_1 \in S_{in}}} (-1)^{a_1+b_1+c_1+x_1y_1z_1} P(a_1b_1c_1|x_1y_1z_1). \tag{77}$$

In what follows, we only consider the case of $a_1 = 0$ and $x_1 = 0$ to prove the following inequality:

$$P(a_1|x_1) \leqslant \frac{1}{2} + \frac{1}{12}\sqrt{72 - \varpi_{es}^2}, \tag{78}$$

which is inspired by a bipartite QKD scheme [29,48]. Note that the right-hand side of inequality (4) is a concave function of ϖ_{es}. Denote $W = \sum_{\substack{a_1,b_1,c_1 \\ x_1,y_1,z_1 \in S_{in}}} (-1)^{a_1+b_1+c_1+x_1y_1z_1} A_{x_1}^{a_1} B_{y_1}^{b_1} C_{z_1}^{c_1}$ as a tripartite quantum operator. Inspired by the semidefinite

programming relaxation of the bipartite scheme [29,48], from Eq. (77) inequality (78) is equivalent to the following operator inequality:

$$A_0^0 \leqslant \frac{1}{2} + \frac{6}{\sqrt{72-\varpi^2}} - \frac{\varpi}{12\sqrt{72-\varpi^2}}W. \tag{79}$$

This operator inequality is then rewritten into

$$\frac{1}{2} + \frac{1}{2}A_0 \leqslant \frac{1}{2} + \frac{6}{\sqrt{72-\varpi^2}} - \frac{\varpi}{12\sqrt{72-\varpi^2}}W, \tag{80}$$

where $A_0 = A_0^0 - A_0^1$ is a Hermitian operator.

Now, it is sufficient to prove the operator inequality (80) for the guessing probability in Eq. (78). Define four operators L_i as follows [29,44]:

$$\begin{aligned} L_1 = &-\frac{\sqrt{\gamma}}{2}A_1 + \frac{\varpi\sqrt{\gamma}}{24}(B_0C_0 - B_1C_1) \\ &+\frac{\varpi}{48\sqrt{\gamma}}(A_1B_0C_0 + A_1B_1C_1), \end{aligned} \tag{81}$$

$$\begin{aligned} L_2 = &\frac{\sqrt{\gamma}}{8} - \frac{\varpi}{24\sqrt{\gamma}}A_0 + \frac{1}{8\sqrt{\gamma}}(B_0C_0 + B_1C_1) \\ &-\frac{\sqrt{\gamma}}{2}(A_0B_0C_0 + A_0B_1C_1) \\ &+\frac{(36-\varpi^2)\sqrt{\gamma}}{144}(A_1B_0C_0 - A_1B_1C_1), \end{aligned} \tag{82}$$

$$\begin{aligned} L_3 = &\frac{\sqrt{\gamma}}{4}(A_0B_0C_0 - A_0B_1C_1) - \frac{1}{8\sqrt{\gamma}}(B_0C_0 - B_1C_1) \\ &-\frac{(36-\varpi^2)\sqrt{\gamma}}{144}(A_1B_0C_0 + A_1B_1C_1), \end{aligned}$$

$$\begin{aligned} L_4 = &\frac{1}{4\sqrt{\gamma}} - \frac{(36-\varpi^2)\sqrt{\gamma}}{72}A_0 - \frac{\sqrt{\gamma}\varpi}{24}(B_0C_0 + B_1C_1) \\ &-\frac{\varpi}{48\sqrt{\gamma}}(A_1B_0C_0 - A_1B_1C_1), \end{aligned} \tag{83}$$

where $\gamma = \frac{3}{\sqrt{72-\varpi^2}}$. It is from a forward evaluation to check that

$$\sum_{i=1}^{4} L_i^\dagger L_i = \frac{6}{\sqrt{72-\varpi^2}} - \frac{1}{2}A_0 + \frac{\varpi}{12\sqrt{72-\varpi^2}}W \tag{84}$$

$$\geqslant 0 \tag{85}$$

for $\varpi < 6\sqrt{2}$. Equation (84) makes use of the equalities $A_x^2 = B_y^2 = C_z^2 = \mathbb{1}$, and $[A_x, B_y] = [A_x, C_z] = [B_y, C_z] = 0$ (commute relations). Inequality (85) follows from the fact that the right-hand side of Eq. (84) is positive semidefinite, i.e., $\sum_{i=1}^4 L_i^\dagger L_i \geqslant 0$. Finally, inequality (80) is obtained from the concavity of its right-had side for ϖ with $\varpi \in [4\sqrt{2}+2, 6\sqrt{2}]$.

Now, we prove asymptotic secret-key rate. The length of the real secret k is lower bounded by $H_{\min}(A|E) - s_p$ (up to terms of order $\sqrt{s}$), where $H_{\min}(A|E)$ denotes the min-entropy [29,50] of A depending on Eve's information, $s_p \approx sH(A|B)$ is used for error correcting [29,50], and $H(A|B) = -\sum_{a,b} p(a,b)\log_2 p(a|b)$ denotes the conditional entropy depending on the average probability with specific outcomes a, b being observed. Hence, it is essential to get the asymptotic secret-key rate R in Eq. (5) by evaluating $H_{\min}(A|E)$ [29]. After the local measurement of A_1, the joint state of A_1 and Eve can be represented by $\rho_{AE} = \sum_{\mathbf{a}} P(\mathbf{a}|\mathbf{x}_r)|\mathbf{a}\rangle\langle\mathbf{a}| \otimes \rho_E^{\mathbf{a}}$, where $\rho_E^{\mathbf{a}}$ denotes the reduced state owned by Eve conditional on A_1's output $\mathbf{a}$. Let $H_{\min}(\mathbf{a}|E)$ be the min-entropy [29,50] of $\mathbf{a}$ depending on Eve's information related to ρ_{AE}.

Now, we estimate the guessing probability of eavesdropper Eve about the raw key $\mathbf{a} = a_1 \cdots a_s$ using s pairs of the triangular network (Fig. 3). Let $P(\mathbf{a}|\mathbf{x}_r, ez)$ denote the probability of $\mathbf{a}$ conditional on Eve's information. The expected optimal guessing probability for Eve is shown as follows:

$$P_{gs}(\mathbf{a}|E) = \max_z \sum_e P(e|z) \max_{\mathbf{a}} P(\mathbf{a}|\mathbf{x}_r, ez). \tag{86}$$

Denote $\rho_{ABC|ez}$ as the initial state when Eve measures z and obtains the outcome e, where $\rho_{ABC|ez}$ satisfies $\rho_{ABC} = \sum_e P(e|z)\rho_{ABC|ez}$. Denote $\mathbf{A}_{\mathbf{x}_r}^{\mathbf{a}} = \prod_{i=1}^s A_{x_r}^{a_i}$ as POVMs for all outputs $\mathbf{a}$ conditional on A_1's inputs $\mathbf{x}_r$. We get that

$$P(\mathbf{a}|\mathbf{x}_r, ez) = \mathrm{tr}\big[\rho_{ABC|ez}^{\otimes s}\mathbf{A}_{\mathbf{x}_r}^{\mathbf{a}}\big], \tag{87}$$

where $\mathbf{x}_r$ denote all parties' inputs with x_r for each experiment. By using the semidefinite programming relaxation [29,48], it follows that

$$\mathbf{A}_{\mathbf{x}_r}^{\mathbf{a}} \leqslant \mathbf{A}(\varpi) = \prod_{i=1}^{s}[\alpha(\varpi)\mathbb{1} + \beta(\varpi)O_i], \tag{88}$$

where $\alpha(\varpi) = \frac{1}{2} + \frac{1}{12}\sqrt{72-\varpi^2} + \frac{\varpi^2}{12\sqrt{72-\varpi^2}}$, $\beta(\varpi) = -\frac{\varpi}{12\sqrt{72-\varpi^2}}$, and $O_i = \sum_{abc,xyz} g_{xyz}^{abc} A_{x_i}^{a_i} B_{y_i}^{b_i} C_{z_i}^{c_i}$. From inequality (88) we obtain that

$$\begin{aligned} P_{gs}(\mathbf{a}|E) &= \max_z \sum_e P(e|z) \max_{\mathbf{a}} \mathrm{tr}\big[\rho_{\mathrm{ABC|ez}}^{\otimes \mathrm{s}}\mathbf{A}_{\mathbf{x}_\mathrm{r}}^{\mathbf{a}}\big] \\ &\leqslant \max_z \sum_e P(e|z) \min_{\varpi} \mathrm{tr}\big[\rho_{ABC|ez}^{\otimes s}\mathbf{A}(\varpi)\big] \\ &\leqslant \min_{\varpi} \mathrm{tr}\big[\rho_{ABC}^{\otimes s}\mathbf{A}(\varpi)\big] \\ &\leqslant \min_{\varpi} \big(\alpha(\varpi) + \beta(\varpi)\varpi_{es} + s_{es}^{-1/4}\big)^s \qquad (89) \\ &\leqslant \left(\frac{1}{2} + \frac{1}{12}\sqrt{72-\varpi_{es}^2} + s_{es}^{-1/4}\right)^s. \qquad (90) \end{aligned}$$

Inequality (89) follows from the result [29,48] by combining systems B and C, where s_{es} denotes the number of the binary key distilled from the raw key. Inequality (90) follows from the inequality

$$\min_{\varpi}\{\alpha(\varpi) + \beta(\varpi)\varpi_{es}\} \leqslant \frac{1}{2} + \frac{1}{12}\sqrt{72-\varpi_{es}^2} \tag{91}$$

using the concavity.

Denote $R := \frac{H_{\min}(\mathbf{a}|E)}{s} = -\frac{\log_2 P_{gs}(\mathbf{a}|E)}{s}$ as the asymptotic secret-key rate of the tripartite QKD scheme [29,50]. It follows from inequality (90) that

$$\begin{aligned} R &= -\frac{\log_2 P_{gs}(\mathbf{a}|E)}{s} \\ &\geqslant -\log_2\left(\frac{1}{2} + \frac{1}{12}\sqrt{72-\varpi_{es}^2}\right) - H(a|b,c), \end{aligned} \tag{92}$$

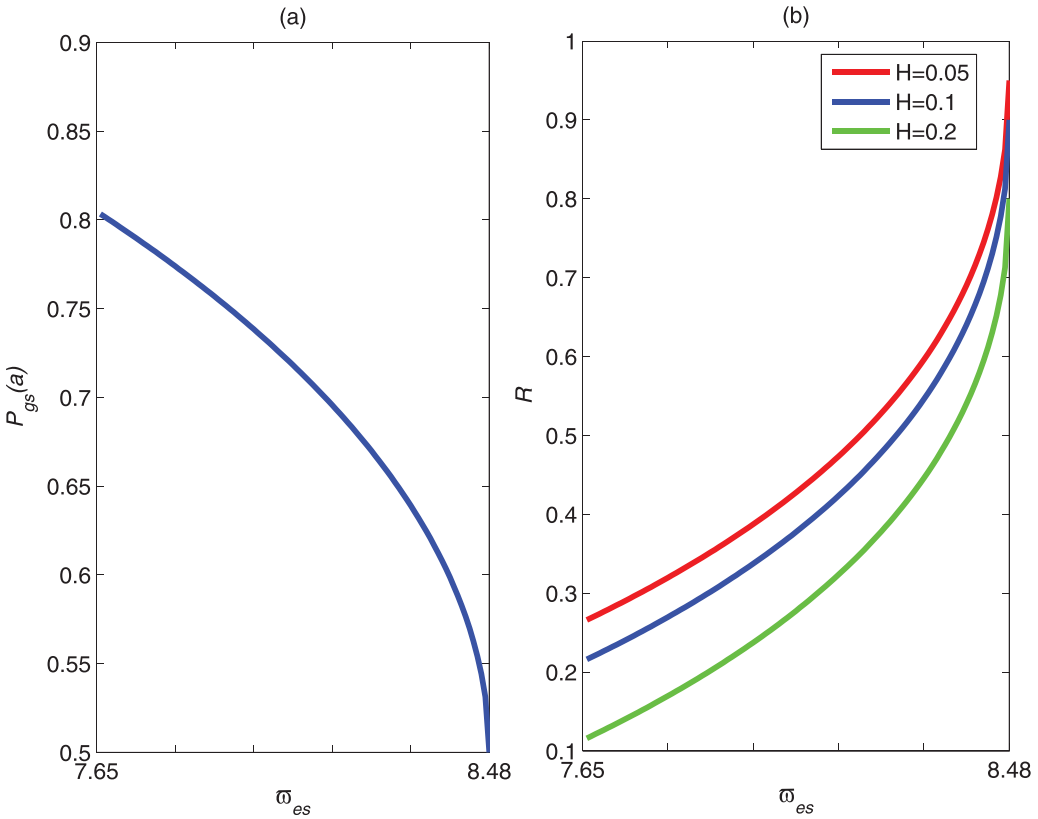


FIG. 8. The numeric simulations of $P_{gs}(\mathbf{a}|E)$ and R. (a) The numeric simulation of $P_{gs}(\mathbf{a}|E)$ via the violation ϖ_{es}. (b) The numeric simulation of R via violation ϖ_{es} and information $H(a|b,c)$ needed for the error correction.

where $P_{gs}(\mathbf{a}|E)$ denotes the expected optimal guessing probability conditional on the eavesdropper's input E. This completes the proof. ■

From Theorem 7, the numeric simulations of bounds of $P_{gs}(\mathbf{a}|E)$ and R are shown in Fig. 8. These bounds are dependent on the practical violation ϖ_{es} obtained in experiments and the information needed for the error correction.

VI. DEVICE-INDEPENDENT BLIND QUANTUM COMPUTATION

Measurement-based quantum computation (MQC) provides a new computation model that makes use of simple measurement on qubits prepared in a highly entangled state [37]. Some examples are shown in Fig. 9 by using specific systems [51,52].

A. Device-independent blind quantum computation

MQC allows one party with limited computational power to use other parties' computational resources without revealing the real task, inputs, and outcomes [53–55]. So-called blind quantum computation (BQC) provides a secure way to verify an adversarial quantum processor utilizing the statistics of local measurements [54,56]. Inspired by a recent scheme [57], a two-server BQC scheme is shown in Fig. 10(a) and a device-independent BQC (DIBQC) scheme is shown in Fig. 10(b):

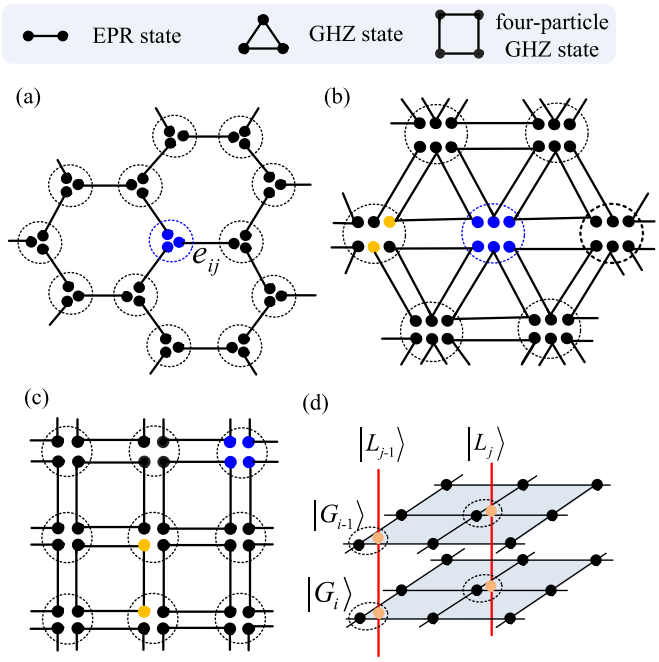


FIG. 9. Quantum networks for measurement-based quantum computation. (a) Honeycomb network consisting of singlets e_{ij}'s [37]. (b) Triangular network consisting of GHZ states. (c) Square network consisting of four-particle GHZ states. (d) Cyclic network consisting of two-dimensional graph states $|G_i\rangle$ and one-dimensional graph states $|L_j\rangle$.

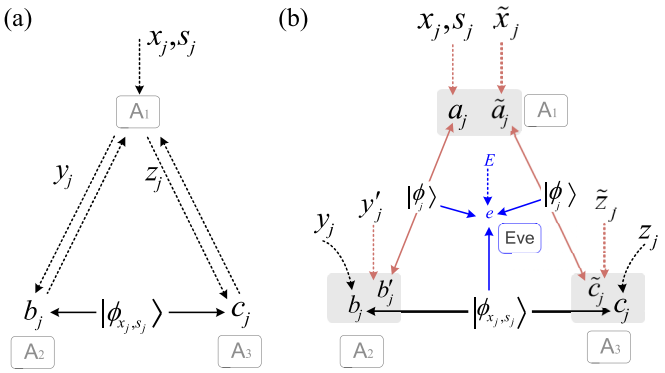


FIG. 10. DIBQC with two servers. (a) A two-server BQC scheme. A_1 delegates a computational task to two servers A_2 and A_3 who share some EPR states of $\otimes_j|\phi_{x_j,s_j}\rangle$. y_j and z_j are bit encodings of the respective θ'_j and $\theta'_j + b_j\pi$. (b) DIBQC with two servers. A_1 and A_2 (or A_3) share a random key k_j (or $\hat{k}_j$) using the shared EPR states (QKD with purple lines). The measurement outcomes y_j, z_j, b_j, and c_j are then encrypted with the one-time pad. An eavesdropper E can correlate its system with the shared sources.

(i) A_1 (verifier) secretly chooses independent strings $\mathbf{x} = x_1 \cdots x_n$ and $\mathbf{s} = s_1 \cdots s_n$ with the uniform probability distribution.

(ii) A_2 and A_3 share some EPR states: $\otimes_{j=1}^m|\phi_{x_j,s_j}\rangle$, where $|\phi_{x_j,s_j}\rangle = \frac{1}{\sqrt{2}}(\mathbb{1}\otimes\sigma_x^{s_j}\sigma_z^{x_j})(|00\rangle + |11\rangle)$ with Pauli operators σ_x and σ_z.

(iii) A_1 shares some EPR states $\otimes_j|\phi_j\rangle$ with A_2 and A_3, respectively.

(iv) A_1 measures the jth entanglement shared with A_2 in the basis $\{\frac{1}{\sqrt{2}}(|0\rangle \pm e^{-i\theta'_j}|1\rangle)\}$ after a local qubit operation $U = |0\rangle\langle 0| + e^{-2i\theta'_j}|1\rangle\langle 1|$, in which $\theta'_j = (-1)^{s_j}\theta_j + x_j\pi$ and θ_j is randomly chosen from $\{\frac{k\pi}{4}|k = 0, 1, \ldots, 7\}$. A_2 measures the jth entanglement shared with A_1 in the basis randomly chosen from $\{\{\frac{1}{\sqrt{2}}(|0\rangle \pm e^{-i\theta'_j}|1\rangle)\}, \forall j\}$. They detect eavesdroppers with public discussions, and then distill a random key $k_1 \cdots k_t$.

(v) A_2 measures the local qubit of the jth Bell state shared with A_3 in the basis $\{\frac{1}{\sqrt{2}}(|0\rangle \pm e^{-i\theta'_j}|1\rangle)\}$, and then sends the outcome $b_j \in \{0, 1\}$ in accord with the one-time pad with $k_1 \cdots k_t$.

(vi) A_1 and A_3 start the modified single-server protocol [53] with the following encoding: $\hat{\theta}_j \mapsto \hat{\theta}'_j + b_j\pi$, $j = 1, 2, \ldots, m$. Here, the modified scheme means that all of A_1's classical information is encoded into the one-time pad using the key which is shared by A_1 and A_3 [54].

The main difference between the two protocols in Fig. 5 is the transmission of classical information. A_1 uses the statistics to verify the shared random key with A_2 or A_3. Hence, the present DIBQC is sensitive to cheating [55,57]. A_3 cannot obtain A_2's outputs from A_1's inputs [56] because the two servers are forbidden to communicate in the present model. Similar to the single-server protocol [55], both servers cannot learn the hidden computational task. From Eq. (74) the guessing probability about A_1's outcome is given by $P_{gs}(\mathbf{a}|E) \leqslant (\frac{1}{2} + \frac{1}{12}\sqrt{72 - \varpi_{es}^2} + m_{es}^{-1/4})^m$, where m_{es} denotes the number of distilled keys. One may further estimate the guessing probability for A_1's computational task [54,55]. Here, A_1 is able to perform quantum measurement while EPR states should be prepared and distributed by the servers [55,57].

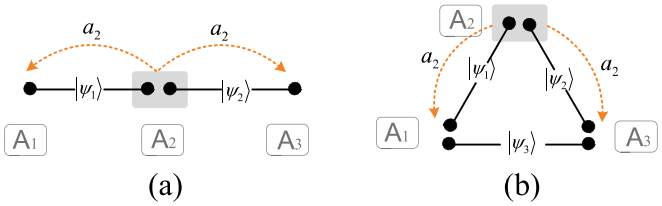


FIG. 11. Entanglement swapping network. (a) An entanglement swapping network consisting of three parties. (b) A triangle network consisting of three parties. $|\psi_i\rangle$ are bipartite entangled pure states.

B. Verifying resources for quantum computation

Inequality (2) provides a useful way to verify these computational resources. Another way is assisted with LOCC as follows.

Take the network shown in Fig. 11 as an example. For Fig. 11(a), each pair of A_1 and A_2, or A_2 and A_3, shares one bipartite entangled pure state $|\psi_i\rangle$. Note that any two parties, A_2 and A_3 for example, can recover a new bipartite entangled state $|\phi_2\rangle$ when the other party A_1 performs a two-particle joint measurement and sends out outcomes. Now, A_2 and A_3 share one generalized bipartite entangled state $|\phi_i\rangle$. It means that any two parties can rebuild a bipartite entangled state assisted by LOCC. Moreover, consider the network as shown in Fig. 11(b), where each pair of A_1 and A_2, A_2 and A_3, and A_3 and A_1 shares one bipartite entangled pure state $|\psi_i\rangle$. Any two adjacent parties (A_2 and A_3 for example) can recover a new bipartite entangled state $|\psi_4\rangle$ when the other (A_1 in example) performs a two-particle joint measurement and sends out outcomes. A_2 and A_3 share two generalized bipartite entangled states $|\psi_2\rangle|\psi_4\rangle$. So, any two parties rebuild a bipartite entangled state assisted by LOCC [58]. This fact is useful for detecting the genuinely multipartite nonlocality of all the multipartite entangled pure states using tailed CHSH inequalities. Interestingly, it is also applicable for network scenarios. Here, the detecting experiment consists of two steps. One is to construct one bipartite entangled state for each pair with LOCC. The other is to verify a generalized bipartite network.

Theorem 8. The genuinely multipartite nonlocality of any connected quantum networks consisting of genuinely entangled multipartite pure states can be verified with the help of LOCC.

Proof. The proof is divided into three steps inspired by a recent result for a single entangled system [58]. The first is to prove that any genuinely multipartite entangled pure states can be useful for entanglement swapping. The second is to prove any two observers in a connected network consisting of genuinely multipartite entangled pure states can recover a new bipartite entangled pure state with the help of others' LOCC. The last is to verify any two parties with shared k bipartite entangled pure states using the maximal violation of some bipartite Bell inequality.

Consider a 4-partite network consisting of Alice, Bob, Charlie, and Tom, where Alice, Bob, and Tom share one genuinely entangled pure state $|\psi_1\rangle$, and Bob and Charlie and Tom share the other genuinely entangled pure state $|\psi_2\rangle$. Alice and Charlie can create a new bipartite entangled pure state with the help of LOCC of Bob and Tom. Now, consider a k-partite connected network consisting of $\mathsf{A}_1, \ldots, \mathsf{A}_k$, where A_i shares at least one genuinely multipartite entangled pure state $|\psi_i\rangle$ with other parties. Any two parties who do not share any entangled state can create a new bipartite entangled pure state with the help of LOCC of other parties.

Now, we continue to prove the result. Consider an n-partite connected network $\mathcal{N}$ consisting of $\otimes_i|\psi_i\rangle$, where $|\psi_i\rangle$ is a genuinely multipartite entangled pure state on Hilbert space $\otimes_{j=1}^{s_i}\mathcal{H}_j$. The proof is similar to that for a single-source network [58]. For any pair of parties A_{i_1} and A_{i_2}, two cases will be discussed. One is that A_{i_1} and A_{i_2} share one entangled pure state [59]. The other is that A_{i_1} and A_{i_2} do not share any entangled pure state.

Step 1. $\mathcal{N}_q$ has genuinely n-partite nonlocality if there exists a covering set such that each pair in it can create entanglement in all branches of the LOCC protocol for other parties.

By using the tilted-CHSH family of Bell inequalities [58,60,61], there is a linear Bell inequality defined by

$$B_{i_1 i_2} = \sum_{\mathbf{s}} \sum_{\mathbf{a},\mathbf{x}} B^{(\mathbf{s})}_{\mathbf{x};\mathbf{a}} \leqslant C \tag{93}$$

when at least one pair of A_{i_1} and A_{i_2} cannot create one bipartite entangled pure state. Here, $B_{i_1 i_2}$ is used to verify the final entangled pure state for A_{i_1} and A_{i_2}, and $B^{(\mathbf{s})}_{\mathbf{x};\mathbf{a}}$ depends on the specific branch $\mathbf{s}$ and inputs $\mathbf{a}$ and outputs $\mathbf{x}$ of all involved parties [58,60,61]. Otherwise, the achievable quantum upper bound is $C' > C$. Here, we choose C, C' to be independent of $i_1 i_2$ [58,60]. Define

$$\mathcal{B} = \sum_{(i_1,i_2)} B_{i_1 i_2}, \tag{94}$$

where the summation is taken over all pairs in a covering set.

Assume that the total state of $\mathcal{N}_q$ is a biseparable state in Eq. (3). From the definition of LOCC, the joint system shared by A_{i_1} and A_{i_2}, which are in different partitions, remains in a separable state in each branch of the protocol [58]. So the maximum of $B_{i_1 i_2}$ is at most C if $i_1 \in I$ and $i_2 \in \bar{I}$. It implies that $\max \mathcal{B} = (k-1)C' + C < kC'$ for any biseparable state ρ_{bs} while $\max \mathcal{B} = kC'$ for a fully entangled $\mathcal{N}_q$.

Step 2. $\mathcal{N}_q$ consisting of generalized EPR states and genuinely entangled GHZ states satisfies the assumption in step 1.

Consider an n-partite connected network $\mathcal{N}_q$ consisting of generalized EPR states $\otimes_{i=1}^{m_1}|\phi_i\rangle$ and genuinely entangled GHZ states $\otimes_{j=1}^{m_2}|\psi_j\rangle$. Since $\mathcal{N}_q$ is connected, $|\Omega\rangle := \otimes_{i=1}^{m_1}\otimes_{j=1}^{m_2}|\phi_i\rangle|\psi_j\rangle$ can be viewed as an n-partite entangled pure state with local dimension d, where d is the maximal dimension of all local systems. Otherwise, there exist at least two parties who cannot share an entangled pure state with LOCC. From Lemmas 1 and 2, it means that two parties are not connected; i.e., the network is disconnected. A simple fact of these networks is as follows: For each A_i, there are projections on local systems such that the remaining network of the other $n-1$ parties also consists of EPR states and GHZ states with LOCC. Here, the measurement outcomes are sent to $n-2$ parties. After these operations, any pair of parties can share at least one generalized EPR state with LOCC of other $n-2$ parties.

Step 3. Perform a Bell test for two parties with shared bipartite entangled pure states $\otimes_i|\phi_i\rangle$.

For a connected network, any pair of parties A_i and A_j can create k_{ij} bipartite entangled pure states $\otimes_{s=1}^{k_{ij}}|\phi_s\rangle$, where $|\phi_s\rangle = \cos\theta_s|00\rangle + \sin\theta_s|11\rangle$. We can rewrite the total state into

$$\otimes_{s=1}^{k_{ij}}|\phi_s\rangle = \sum_{t=0}^{2^k-1}\alpha_t|tt\rangle, \tag{95}$$

where $\sum_{t=0}^{2^{k_{ij}}-1}\alpha_t|tt\rangle =: |G\rangle$ is a bipartite state on Hilbert space $\mathbb{C}^{2^{k_{ij}}} \times \mathbb{C}^{2^{k_{ij}}}$, and α_t's are coefficients depending on θ_s's. There is a tailored Bell inequality which will be maximally violated by this state. One example is the tailed CHSH [60] for verifying the nonlocality of $|G\rangle$ for some $\theta_i \neq 0$. Another method is using a linear Bell inequality with three and four inputs and $2^{k_{ij}}$ outputs [61].

Step 4. $\mathcal{N}_q$ consisting of all genuinely entangled pure states satisfies the assumption in step 1.

For each pair of A_i and A_j, there exists a generalized chain-shaped subnetwork consisting of $\mathsf{A}_i, \mathsf{A}_{s_1}, \ldots, \mathsf{A}_{s_\ell}, \mathsf{A}_j$. For each entangled pure state $\rho_{1\cdots m}$, there exist local measurements on the mth particle such that the resultant is entangled pure state $\varrho_{1\cdots m-1}$ [58]. By using these local operations for all the other parties (iteratively for each multipartite state), they can recover a standard chain-shaped subnetwork consisting of all d-dimensional bipartite entangled pure states $|\Phi_j\rangle$. Similar to Lemma S3, A_i and A_j can recover at least one bipartite entangled pure state using the local d-dimensional rotations after $\mathsf{A}_{s_1}, \ldots, \mathsf{A}_{s_\ell}$ perform local joint measurement under the generalized Bell basis: $\{|B_{st}\rangle = \frac{1}{\sqrt{d}}\sum_{j=0}^{d-1}\exp(\frac{2is j\pi}{d})|j\rangle|d-j\rangle\}$. Finally, the nonlocality of the resultant can be verified by A_i and A_j using the Bell inequality with multiple setting [61]. This completes the proof. ■

Theorem 9. Consider a connected n-partite network $\mathcal{N}_q$ consisting of EPR and GHZ states with white noise. $\mathcal{N}_q$ has genuinely n-partite nonlocality with the help of LOCC if each noisy state can be verified with a Bell inequality.

The proof is shown in Appendix C.

Example 3. A useful application of these results is to verify quantum resources for measure-based quantum computation, as shown in Fig. 4. In particular, consider universal resources of hypergraph states $|E\rangle = \prod_{\{i_1,\cdots,i_k\}\in E} CZ_{i_1 i_2 \cdots i_k}|+\rangle^{\otimes n}$, where $\{i_1, \ldots, i_k\}$ denote k vertices connected by a k-hyperedge in edge set E [50], and $CZ_{i_1\cdots i_k}$ are controlled-phase gates among the k connected qubits. One example is the W-type state $\frac{1}{2}(|000\rangle + |010\rangle + |100\rangle + |111\rangle)$ from a 3-hyperedge that is inequivalent to the GHZ state from a chain-shaped graph under LOCC. The genuinely multiparticle nonlocality of a single hypergraph state is verified using Hardy-type arguments or Bell inequalities [62]. Theorem 9 proves the genuinely multipartite nonlocality of connected networks consisting of graph states and hypergraph states. It provides a general witness of

hybrid resources for measurement-based quantum computation.

VII. DEVICE-INDEPENDENT QUANTUM SECRET SHARING

In an (n, k) secret sharing scheme, a secret x is split into n parts $x_1, x_2, \ldots, x_n$ such that x is recoverable from any k parts [63,64]. Inspired by a recent scheme with the GHZ state [65], our goal is to propose a device-independent quantum secret sharing (DIQSS) using quantum networks based on inequality (2). For simplicity, a (3,2) DIQSS is constructed using the triangular network $\mathcal{N}_q$ in Fig. 7(a). For a secret bit s, the goal is to distribute the split parts s_1, s_2, and s_3 to A_1, A_2, and A_3, respectively, where $s_i \oplus s_j = s$ with $1 \leqslant i \neq j \leqslant 3$. In the Bell model (Fig. 7), suppose that $a_1 = s_1$, $a_2 = s_2$, and $b_2 = s_3$ for specified inputs x_i, y_i, and z_i. This implies that any two parties can cooperate to recover the secret s, but one party fails. The violation of the Mermin inequality [41] can rule out the outer eavesdropper but not the inner untrustful party. The main reason is that a local correlating operation of a receiver is equivalent to a measurement-dependent hidden variable model [65]. Suppose that A_i is an eavesdropper who recovers the output a_1 of A_1 with $i \neq 1$. The guessing probability $P(e = a_1|\mathbf{x}, u)$ by eavesdroppers can be reshaped using the conditional probability $P(\mathbf{a}, e|\mathbf{x}, u)$ into a linear optimization as

$$\max_{P(\mathbf{a},e|\mathbf{x},u)} P(e = a_1|\mathbf{x}, u) \tag{96}$$

$$\text{such that} \quad \sum_{\mathbf{a},e} P(\mathbf{a}, e|\mathbf{x}, u) = 1, \forall \mathbf{x}, u, \tag{97}$$

$$P(\mathbf{a}, e|\mathbf{x}, u) \geqslant 0, \forall \mathbf{a}, \mathbf{x}, e, u, \tag{98}$$

$$\sum_{w} (P(\mathbf{a}, e|\mathbf{x}, u) - P(\mathbf{a}, e|\mathbf{x}^{\zeta}, u)) = 0, \quad \forall \zeta, \tag{99}$$

$$P(\mathbf{a}|\mathbf{x}) = \sum_{u,e} p(u) P(\mathbf{a}, e|\mathbf{x}, u), \tag{100}$$

$$\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} \mathcal{L}_i(P(\mathbf{a}|\mathbf{x})) > 2^n\sqrt{2} - 4\sqrt{2} + 2, \tag{101}$$

where $\mathbf{a} = a_1 \cdots a_n$, ζ is a subset of $\{a_1, \ldots, a_n\}$, and $\mathbf{x}^{\zeta} = x_1^{\zeta} \cdots x_n^{\zeta}$ satisfies $x_i^{\zeta} = x_i$ with $a_i \notin \zeta$. Equation (99) is a nonsignaling condition for any subset ζ with respect to different inputs of the corresponding parties. Equations (100) and (101) are key to estimate the guessing probability of $P(e = a_1|\mathbf{x}, u)$, where the projected distribution $P(\mathbf{a}|\mathbf{x})$ is the n-variable probability derived from local measurements on $\mathcal{N}_q$. The linear optimization problem is algorithmically solvable using the semidefinite program [66] and Navascues-Pironio-Acín (NPA) hierarchy [44,65].

Example 4. Consider the triangle network shown in Fig. 3 shared by Alice, Bob, and Charlie. Our consideration is to use the triangle network with the present inequality (2). Here, inequality (19) given by

$$\begin{aligned}\varpi &= A_1B_1C_1 + A_2B_1C_1 + A_1B_2C_2 - A_2B_2C_2 \\ &\quad + A_3B_3C_1 + A_3B_4C_1 + A_4B_3C_2 - A_4B_4C_2 \\ &\quad + A_3B_5C_3 + A_3B_5C_4 + A_4B_6C_3 - A_4B_6C_4 \\ &\leqslant 4\sqrt{2} + 2\end{aligned} \tag{102}$$

will be used to construct a (3,2) DIQSS. Here, assume that Alice is untrusted and tries to recover the output of other parties. Note that the maximal violation of this inequality is given by $\Delta = 6\sqrt{2}$, that is,

$$\begin{aligned}A_1B_1C_1 + A_2B_1C_1 + A_1B_2C_2 - A_2B_2C_2 &= 2\sqrt{2}, \\ A_3B_3C_1 + A_3B_4C_1 + A_4B_3C_2 - A_4B_4C_2 &= 2\sqrt{2}, \\ A_3B_5C_3 + A_3B_5C_4 + A_4B_6C_3 - A_4B_6C_4 &= 2\sqrt{2}.\end{aligned}$$

This implies that Bob and Charlie are maximally entangled for quantum settings or share a Popescu-Rohrlich (PR) box in the nonsignaling settings even if they know the input and output of Alice. Hence, Bob and Charlie can ensure that their reduced states are uncorrelated with any other system if they can get the maximal violation of inequality (102). In this case, the guess probability of the output a_2 or a_3 is given by $p(e = a_2|x_1, x_2, x_3) = p(e = a_3|x_1, x_2, x_3) = 1/2$, that is, a random guess. This means that the maximal violation rules out the guess attack of an untrusted party. In applications, the secure scheme can be obtained with large violation. The main idea is solving the linear optimization given in Eqs. (6)– (11) by using the NPA hierarchy [44,65]. Eve's guess probability $p(e = c|x_1, x_2, x_3)$ in quantum settings is given by $p(e = c|x_1, x_2, x_3) \leqslant 3/4$ for $\varpi \geqslant 8.25$ and $P_{\text{guess}} = 1/2$ for $\varpi = 6\sqrt{2}$. Note that inequality (102) is not symmetric. This implies a larger violation $\varpi \geqslant 8.31$ required for $p(e = b|x_1, x_2, x_3) \leqslant 3/4$ for guessing the output b. These are larger than the recent scheme [65] using a GHZ state and Svetlichny inequality with fewer inputs [33]. Generally, perfect security (in terms of random guess) can be ensured by the maximal violation of the present inequality. This provides a useful method for featuring the quantum secret sharing. Similar optimization problems can be formed for parties being eavesdroppers.

VIII. CONCLUSION

The multipartite correlations generated by a k-independent network form a star-convex set [20]. There is a convex correlation set with relaxed input assumption arising from inequality (2). Compared with previous nonlinear Bell inequalities [15,18–20], inequality (2) enables a standard Bell test without assumptions of source independence for verifying general quantum networks, including acyclic networks [20], cyclic networks [34], and universal computational resources for measurement-based quantum computation [51–53]. The main drawback is the lower visibility for noisy networks. It may be applicable when both the number of parties and the number of entangled sources are not large enough. This remains an interesting problem for further improvement.

The independence of sources cannot be guaranteed in network scenarios because any untrusted party may correlate the shared independent sources into a new one. This is essentially different from the independence assumption of measurement settings [67], which is resolvable both in theory [68] and experimental improvements [69,70]. The present inequality (2) allows remote legitimate parties to share a common random key in a fully device-independent manner. Another example is device-independent BQC [55,56]. The last example is sharing of a classical secret [44]. The formal security proofs of

these applications and other candidates should be of interest in quantum information processing using practical quantum resources.

ACKNOWLEDGMENTS

We thank Ronald de Wolf, Carlos Palazuelos, Luming Duan, Yaoyun Shi, M. Orgun, J. Pieprzyk, Yuan Su, Huiming Li, Xiubo Chen, and Yixian Yang for their help. This work was supported by the National Natural Science Foundation of China (Grants No. 61772437 and No. 62172341), Sichuan Youth Science and Technique Foundation (Grant No. 2017JQ0048), and Fundamental Research Funds for the Central Universities (Grant No. 2018GF07).

APPENDIX A: PROOF OF LEMMA 1

By representing an EPR state as one edge with two nodes, $\mathcal{N}_q$ can be schematically represented by a connected graph $\mathcal{G}$ in which each party owns some nodes of edges. Here, a connected graph means that for any two nodes, there is a path connecting them. For each pair of two parties who share EPR states, they are connected by edges. The result is easily followed for an acyclic graph which has no cycle, that is, a tree graph $\mathcal{G}$ [38]. In fact, note that the total number of particles is even. The total number of parties who have an odd number of particles is even. It means that the total number of parties who have an even number of particles is even. Consider one pair A_i and A_j that has even number of particles. Since $\mathcal{G}$ is connected, there is a connected subgraph $\mathcal{G}_{i\to j} = \{\mathsf{A}_i, \mathsf{A}_{s_1}, \ldots, \mathsf{A}_{s_k}, \mathsf{A}_j\}$, where $\mathsf{A}_{s_1}, \ldots, \mathsf{A}_{s_k}$ have an odd number of particles. In this case, each adjacent pair of $\mathcal{G}_{i\to j}$ can transform one shared EPR state $|\phi_i\rangle|0\rangle|0\rangle$ into a four-particle GHZ state $|\hat{\phi}_i\rangle = \cos\theta_i|0000\rangle + \sin\theta_i|1111\rangle$ by using a local controlled-NOT operation and one auxiliary particle in the state $|0\rangle$ for each party. After these local operations are performed, all the parties in $\mathcal{G}_{i\to j}$ have an odd number of particles. This procedure can be iteratively completed for all parties who have an even number of particles.

In what follows, we propose Algorithm 1 to locally transform a cyclic graph. Here, a cyclic graph $\mathcal{G}$ means that it contains at least one cycle.

For $\mathcal{G}_\ell$, each vertex has only added an even number of nodes, even if local transformations may change the number of nodes for these vertices contained in $\mathcal{G}_j$ for $j < \ell$. In this case, all the vertices $\mathsf{B}_s \in \mathcal{G}_j$ have odd numbers of nodes after local transformations being performed by all vertices in $\mathcal{G}_\ell$. It means that one does not need to consider all the subgraphs $\mathcal{G}_j$ with $j < \ell$ after $\mathcal{G}_\ell$ has been changed. This is key to stopping Algorithm 1.

We take use of this algorithm to prove Lemma 1. Note that n is even and each edge has two nodes. Moreover, two nodes will be added each time. For each round, there are two vertices A_i and A_j being relabeled as B_i and B_j while other vertices B_s are unchanged. By induction on the number n of vertices, all the vertices A_j can be relabeled as B_j because the network $\mathcal{N}_q$ is connected. This can be proved by contradiction. In fact, assume that one vertex has an even number of particles while $n-1$ vertices have odd numbers of nodes. There are odd nodes because $n-1$ is an odd integer. This contradicts the fact that there is an even number of vertices from two nodes of each edge. This completes the proof of Lemma 1. ■

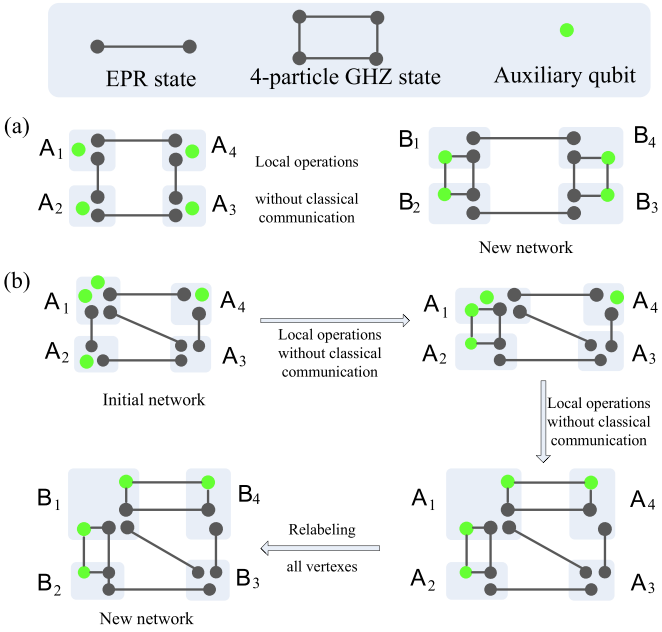


FIG. 12. Schematically transforming a network using local operations and auxiliary particles without classical communication. (a) Cyclic network with four parties. Four parties $\mathsf{A}_1, \ldots, \mathsf{A}_4$ share four EPR states. (b) Cyclic network with two cycles. Four parties $\mathsf{A}_1, \ldots, \mathsf{A}_4$ share five EPR states.

Example 3. Two networks are shown in Fig. 12. In Fig. 12(a), there are four parties who share four EPR states. In step S2, A_1 and A_2 can locally change an EPR state into a four-particle GHZ state with local operations and auxiliary particles. A similar result holds for A_3 and A_4. After these local operations are performed, they obtain a new network consisting of two EPR states and two four-particle GHZ states, where each party has three particles. In Fig. 12(b), there are four parties who share five EPR states. Different from the network shown in Fig. 12(a), there are two cycles, that is, $\mathcal{G}_1 = \{\mathsf{A}_1, \mathsf{A}_2, \mathsf{A}_3\}$ and $\mathcal{G}_2 = \{\mathsf{A}_1, \mathsf{A}_3, \mathsf{A}_4\}$. In step S2, consider $\mathcal{G}_1$ first. A_1 and A_2 can change an EPR state into a four-particle GHZ state with local operations and auxiliary particles. And then, consider $\mathcal{G}_2$. A_1 has four particles by adding the particles involved in cycle $\mathcal{G}_1$. So, it is sufficient to consider the EPR state shared by A_1. Specifically, A_1 and A_4 transform the shared EPR state into a four-particle GHZ state with local operations and auxiliary particles. After these local operations, they obtain a new network consisting of three EPR states and two four-particle GHZ states, where each party has an odd number of particles.

APPENDIX B: PROOF OF THEOREM 6

1. Case 1: Generalized entanglement swapping on any connected acyclic quantum networks

Consider a connected acyclic quantum network $\mathcal{N}_q$ consisting of EPR states $|\phi_i\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, as shown in Fig. 6(a). One goal of $\mathcal{N}_q$ is that all parties $\mathsf{A}_1, \ldots, \mathsf{A}_k$ who have no prior shared entanglement create a multipartite

GHZ state with the help of other parties $\mathsf{A}_{k+1}, \ldots, \mathsf{A}_n$. In experiment, each party A_{k+j} performs a multiparticle Bell measurement on the local system and sends the outcome for other parties A_i, who can recover a k-particle GHZ state $|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes k} + |1\rangle^{\otimes k})$ by performing proper local unitary operations. Suppose that all parties A_i finally obtain binary outcomes $a_i \in \{-1, 1\}$ after performing local qubit measurements. Here, A_i's perform the single-qubit measurement with measurement input $\vec{x}_i = (\cos\theta_i, 0, \sin\theta_i)$ on a Bloch sphere. Their outcomes have correlations as follows:

$$\begin{aligned} E_q(\vec{\mathbf{x}}) &= \sum_{\Xi_1} P_q(\mathbf{a}|\vec{\mathbf{x}}) - \sum_{\Xi_2} P_q(\mathbf{a}|\vec{\mathbf{x}}) \\ &= \prod_{i=1}^{k} \cos\theta_i + \prod_{i=1}^{k} \sin\theta_i, \end{aligned} \tag{B1}$$

where $\vec{\mathbf{x}} = \vec{x}_1 \cdots \vec{x}_k$ are measurement vectors under the Pauli basis of a single-qubit system.

In what follows, we show that these correlations can be simulated classically, as shown in Fig. 6(b) with even k. A similar proof holds for odd k. The classical simulation is completed by powerful eavesdroppers E_i who can access all the measurement inputs with fully independent variables λ_j. The proof is completed by the following separable steps:

(S1) Eavesdroppers divide the simulation network shown in Fig. 6(b) into n chain-shaped subnetworks $\mathcal{N}_1, \ldots, \mathcal{N}_k$ with the help of classical communication, where each subnetwork $\mathcal{N}_i$ contains E_i and E_{k+j}, and all subnetworks are connected and $\mathcal{N}_i \cap \mathcal{N}_j \neq \varnothing$.

(S2) For the subnetwork $\mathcal{N}_1$, eavesdroppers set up the first simulation according to finite classical communication. They obtain a classical correlation as

$$\begin{aligned} E_c^1 &= \sum_{\Xi_1} P_c(\mathbf{e}_1|\vec{\mathbf{x}}_1) - \sum_{\Xi_2} P_c(\mathbf{e}_1|\vec{\mathbf{x}}_1) \\ &= 2\cos\theta_1 \end{aligned} \tag{B2}$$

from the multipartite classical simulation protocol given in Sec. IV, where $\vec{\mathbf{x}}_1$ and $\mathbf{e}_1$ denote the respective inputs and outcomes of all eavesdroppers included in the subnetwork $\mathcal{N}_1$, the input of the eavesdropper E_1 is given by $\vec{x}_1' = (2\cos\theta_1, 0, 0)$, while the inputs of other eavesdroppers are given by $(1, 0, 0)$. Note that this simulation procedure does not require to correlate variables λ_i.

(S3) By repeating step S2 for all subnetworks $\mathcal{N}_i$, eavesdroppers can obtain a classical correlation as

$$\begin{aligned} E_c^i &= \sum_{\Xi_1} P_c(\mathbf{e}_i|\vec{\mathbf{x}}_i) - \sum_{\Xi_2} P_c(\mathbf{e}_i|\vec{\mathbf{x}}_i) \\ &= \cos\theta_i \end{aligned} \tag{B3}$$

for $i = 2, \ldots, n$, where $\vec{\mathbf{x}}_i$ and $\mathbf{e}_i$ denote the respective inputs and outcomes of eavesdroppers included in the subnetwork $\mathcal{N}_i$, the input of the eavesdropper E_i is given by $\vec{x}_i' = (\cos\theta_i, 0, 0)$, while the inputs of all the other eavesdroppers are given by $(1, 0, 0)$ in the subnetwork $\mathcal{N}_i$. Note that these steps are independent. It implies that

$$\begin{aligned} E_c(\vec{\mathbf{x}}') &= \sum_{\Xi_1} P_c(\mathbf{e}|\vec{\mathbf{x}}') - \sum_{\Xi_2} P_c(\mathbf{e}|\vec{\mathbf{x}}') \\ &= \prod_{i=1}^{n} E_c^i \\ &= 2\prod_{i=1}^{k} \cos\theta_i, \end{aligned} \tag{B4}$$

which can be followed from the Bell observables of A_{k+j}s shown in Fig. 6(b), where $\vec{\mathbf{x}}'$ and $\mathbf{e}$ denote the respective inputs and outputs of eavesdroppers included in $\mathcal{N}_q$.

(S4) Eavesdroppers set up the second classical simulation with the help of finite classical communication. They obtain multipartite correlations

$$\begin{aligned} \hat{E}_c^i &= \sum_{\Xi_1} P_c(\mathbf{e}_i'|\vec{\mathbf{x}}_i'') - \sum_{\Xi_2} P_c(\mathbf{e}_i'|\vec{\mathbf{x}}_i'') \\ &= \sin\theta_i \end{aligned} \tag{B5}$$

for $i \geqslant 2$ and $2\sin\theta_1$ for $i = 1$ from the classical simulation protocol given in Sec. IV, where the input of E_i is given by $\vec{x}_i'' = (0, 0, \sin\theta_i)$ for $i \geqslant 2$ or $\vec{x}_1'' = (0, 0, 2\sin\theta_1)$ while the inputs of other eavesdroppers are given by $(0, 0, 1)$ in the subnetwork $\mathcal{N}_i$, $i = 1, \ldots, n$. Note that these steps are independent. It follows that

$$\begin{aligned} \hat{E}_c(\vec{\mathbf{x}}'') &= \prod_{i=1}^{k} \hat{E}_c^i \\ &= 2\prod_{i=1}^{k} \sin\theta_i. \end{aligned} \tag{B6}$$

Finally, from Eqs. (B4) and (B6) it is implied that

$$\begin{aligned} \sum_{\Xi_1} P_c(\mathbf{e}|\vec{\mathbf{x}}) - \sum_{\Xi_2} P_c(\mathbf{e}|\vec{\mathbf{x}}) &:= \frac{1}{2}(\hat{E}_c(\vec{\mathbf{x}}') + \hat{E}_c(\vec{\mathbf{x}}'')) \\ &= \prod_{i=1}^{k} \cos\theta_i + \prod_{i=1}^{k} \sin\theta_i \\ &= E_q(\vec{\mathbf{x}}) \end{aligned} \tag{B7}$$

from Eq. (B1), where two independent simulations are recombined into one simulation with equal probability. A similar result holds for generalized measurement in terms of the Pauli basis [46,47].

2. Case 2: Entanglement swapping on acyclic networks consisting of generalized EPR states

Consider an n-partite connected acyclic quantum network $\mathcal{N}_q$ consisting of generalized EPR states $|\psi_i\rangle = \alpha_i|00\rangle + \sqrt{1-\alpha_i^2}|11\rangle$, as shown in Fig. 6(a). One goal of $\mathcal{N}_q$ is to help all parties A_i to build a generalized multipartite GHZ state assisted by local measurements of other parties A_{k+j}, $i = 1, \ldots, k$; $j = 1, \ldots, n-k$. In experiment, each party A_{k+j} performs a multiparticle Bell measurement on the local systems and sends the outcome to A_i's who can

recover a generalized k-partite GHZ state $|\mathrm{GHZ}\rangle = \gamma|0\rangle^{\otimes k} + \sqrt{1-\gamma^2}|1\rangle^{\otimes k}$ with success probability p by performing proper local unitary operations, where p and γ depend on α_i and β_i. Note that the probability distribution p can be generated by one party in a classical simulation experiment. So, it is sufficient to prove the result for one generalized GHZ state $|\mathrm{GHZ}\rangle$. Suppose that A_i's obtain binary outcomes $a_i \in \{-1, 1\}$. In the classical simulation, A_i's finally perform the single-qubit measurement with measurement input $\vec{x}_i = (\cos\theta_i, 0, \sin\theta_i)$. Their outcomes for even k exhibit nonlocal correlations of the following form:

$$E_q(\vec{\mathbf{x}}) = \sum_{\Xi_1} P_q(\mathbf{a}|\vec{\mathbf{x}}) - \sum_{\Xi_2} P_q(\mathbf{a}|\vec{\mathbf{x}}) = 2\gamma\sqrt{1-\gamma^2}\prod_{i=1}^{k}\cos\theta_i + \prod_{i=1}^{k}\sin\theta_i. \quad \text{(B8)}$$

Consider a classical network $\mathcal{N}$ consisting of m variables λ_i with the similar network configuration as $\mathcal{N}_q$. Suppose that $\mathcal{N}$ is shared by classical eavesdroppers E_i. For classical simulation, eavesdroppers can perform a probabilistic simulation with an input $(2\gamma\sqrt{1-\gamma^2}, 0, 0)$. From a similar procedure of case 1, it is easy to prove that $E(\vec{\mathbf{x}})$ given in Eq. (B8) can be simulated classically using several independent simulations by eavesdroppers who can access all measurement inputs.

Before closing the proof, we show that step S1 in case 1 can be completed with finite classical communication, whereas other steps require finite classical communication from case 2. Note that $\mathcal{N}$ is acyclic [38]. There is one node that is connected to each agent E_i with only one path. This path is contained in the subnetwork $\mathcal{N}_i$. In theory, an acyclic graph is equivalent to a tree where the root is the desired center node, and some edges are allowed to be in different paths; i.e., one random variable can be used more than one time in classical simulation. Some examples are shown in Fig. 13(a). Generally, a similar result holds for special cyclic networks shown in Figs. 13(b) and 13(c), where all the independent agents are not included in any cycle. This completes the proof.

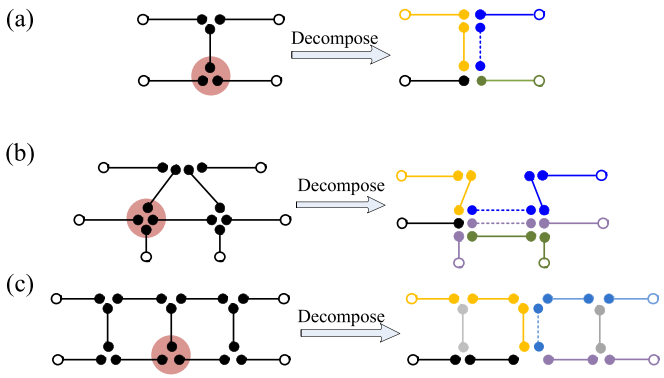


FIG. 13. Decomposing simulation networks. (a) An acyclic network consisting of five random variables. (b) A network with one cycle consisting of nine random variables. (c) A network with two cycles consisting of 11 random variables. Each variable is represented by two dots (or one dot and one circle) connected by one line. All the empty circles denote independent observers who want to create an entanglement in terms of the quantum network. The shadowed node denotes the root of the equivalent tree. The right-hand side of each subnetwork consists of several decomposed simulation networks with several chain-shaped subnetworks, where each subnetwork is represented by one colored path. The dotted lines denote the reused random variables. The grey figures are useless for generalized entanglement swapping experiment. The proof holds for special cyclic networks shown in Figs. 6(b) and 6(c), where all the independent agents are not included in any cycle.

APPENDIX C: PROOF OF THEOREM 9

Proof. First, consider the entanglement swapping network consisting of Alice, Bob, and Charlie, who share two Werner states [40]:

$$\rho_i = v_i|\phi\rangle\langle\phi| + \frac{1-v_i}{4}\mathbb{1}_4, \quad i = 1, 2, \quad \text{(C1)}$$

where $|\phi\rangle$ is an EPR state, $\mathbb{1}_4$ is the identity operator with rank 4, and $v_i \in (0, 1]$. In what follows, we consider the resultant of Alice and Charlie with the help of LOCC of Bob. Specially, Bob performs a two-qubit joint measurement under a Bell basis, $\{\frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)\}$, and sends the outcome to Alice or Charlie, who can then recover a bipartite state as follows:

$$\rho = v_1v_2|\phi\rangle\langle\phi| + \frac{v_1+v_2-2v_1v_2}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|) + \frac{(1-v_1)(1-v_2)}{4}\mathbb{1}_4. \quad \text{(C2)}$$

The fidelity of ρ with respect to the EPR state $|\phi\rangle$ is given by

$$F(\rho, |\phi\rangle\langle\phi|) = \mathrm{tr}[|\phi\rangle\langle\phi|\rho] = \tfrac{v_1+v_2}{2}, \quad \text{(C3)}$$

which satisfies $\min\{v_1, v_2\} \leqslant \frac{v_1+v_2}{2} \leqslant \max\{v_1, v_2\}$. It means that ρ can be verified when both ρ_1 and ρ_2 can be verified using a proper Bell test.

Moreover, consider a k-particle GHZ state with white noise as follows [40]:

$$\rho_{1\cdots k} = v|\psi\rangle\langle\psi| + \frac{1-v}{2^k}\mathbb{1}_{2^k}, \quad \text{(C4)}$$

which is shared by k parties, where $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes k} + |1\rangle^{\otimes k})$, and $\mathbb{1}_{2^k}$ is the identity operator with rank 2^k, and $v \in (0, 1]$. One party performs local measurement under the basis $\{\frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)\}$ and sends the outcome to the others who can recover a $(k-1)$-particle GHZ state with white noise, i.e., $\rho_{1\cdots k-1} = v|\psi'\rangle\langle\psi'| + \frac{1-v}{2^{k-1}}\mathbb{1}_{2^{k-1}}$, where $|\psi'\rangle$ is a GHZ state with $k-1$ particles given by $|\psi'\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes k-1} + |1\rangle^{\otimes k-1})$, and $\mathbb{1}_{2^{k-1}}$ is the identity operator with rank 2^{k-1}.

With two facts stated above, we can prove the result. The only difference for quantum networks consisting of pure states is that the final bipartite states are noisy EPR states $\varrho_1, \ldots, \varrho_k$. Assume that $\rho_1, \ldots, \rho_m$ are input noisy states, where $v_1, \ldots, v_m$ are entanglement fractions. Since each noisy state ρ_i can be verified using some Bell inequality $\mathcal{L}_i \leqslant 0$, $\otimes_j\varrho_j$ can be verified similarly using the Bell inequality $\mathcal{B}_t \leqslant 0$, where t satisfies $v_t = \min\{v_1, \ldots, v_m\}$ from the first fact given above. This completes the proof.

[1] J. S. Bell, On the Einstein-Podolsky-Rosen paradox, Phys. Phys. Fiz. **1**, 195 (1964).

[2] A. Einstein, B. Podolsky, and N. Rosen, Can quantum-mechanical description of physical reality be considered complete? Phys. Rev. **47**, 777 (1935).

[3] L. Masanes, All Bipartite Entangled States Are Useful for Information Processing, Phys. Rev. Lett. **96**, 150501 (2006).

[4] N. Brunner, D. Cavalcanti, S. Pironio, V. Scarani, and S. Wehner, Bell nonlocality, Rev. Mod. Phys. **86**, 419 (2014).

[5] A. K. Ekert, Quantum Cryptography Based on Bell's Theorem, Phys. Rev. Lett. **67**, 661 (1991).

[6] D. Mayers and A. Yao, Quantum cryptography with imperfect apparatus, *Proceedings of the 39th Annual Symposium on the Foundations of Computer Science (FOCS)* (IEEE, Palo Alto, USA, 1998), pp. 503–512.

[7] J. Barrett, L. Hardy, and A. Kent, No Signaling and Quantum Key Distribution, Phys. Rev. Lett. **95**, 010503 (2005).

[8] U. Vazirani and T. Vidick, Fully Device-Independent Quantum Key Distribution, Phys. Rev. Lett. **113**, 140501 (2014).

[9] S. Pironio, A. Acín, S. Massar, A. Boyer de la Giroday, D. N. Matsukevich, P. Maunz, S. Olmschenk, D. Hayes, L. Luo, T. A. Manning, and C. Monroe, Random numbers certified by Bell's theorem, Nature (London) **464**, 1021 (2010).

[10] R. Colbeck and R. Renner, Free randomness can be amplified, Nat. Phys. **8**, 450 (2012).

[11] R. Gallego, L. Masanes, G. D. L. Torre, C. Dhara, L. Aolita, and A. Acín, Full randomness from arbitrarily deterministic events, Nat. Commun. **4**, 2654 (2013).

[12] L.-M. Duan, M. Lukin, J. I. Cirac, and P. Zoller, Long-distance quantum communication with atomic ensembles and linear optics, Nature (London) **414**, 413 (2001).

[13] H. J. Kimble, The quantum Internet, Nature (London) **453**, 1023 (2008).

[14] C. M. Lee and R. W. Spekkens, Causal inference via algebraic geometry: Feasibility tests for functional causal structures with two binary observed variables, J. Causal Inference **5**, 20160013 (2017).

[15] C. Branciard, N. Gisin, and S. Pironio, Characterizing the Nonlocal Correlations Created via Entanglement Swapping, Phys. Rev. Lett. **104**, 170401 (2010).

[16] T. Fritz, Beyond Bell's theorem: Correlation scenarios, New J. Phys. **14**, 103001 (2012).

[17] A. Tavakoli, P. Skrzypczyk, D. Cavalcanti, and A. Acín, Nonlocal correlations in the star-network configuration, Phys. Rev. A **90**, 062109 (2014).

[18] D. Rosset, C. Branciard, T. J. Barnea, G. Pütz, N. Brunner, and N. Gisin, Nonlinear Bell Inequalities Tailored for Quantum Networks, Phys. Rev. Lett. **116**, 010403 (2016).

[19] R. Chaves, Polynomial Bell Inequalities, Phys. Rev. Lett. **116**, 010402 (2016).

[20] M.-X. Luo, Computationally Efficient Nonlinear Bell Inequalities for Quantum Networks, Phys. Rev. Lett. **120**, 140402 (2018).

[21] M. X. Luo, A nonlocal game for witnessing quantum networks, npj Quantum Inf. **5**, 91 (2019).

[22] N. Gisin, J.-D. Bancal, Y. Cai, P. Remy, A. Tavakoli, E. Zambrini Cruzeiro, S. Popescu, and N. Brunner, Constraints on nonlocality in networks from no-signaling and independence, Nat. Commun. **11**, 2378 (2020).

[23] A. Tavakoli, Bell-type inequalities for arbitrary noncyclic networks, Phys. Rev. A **93**, 030101(R) (2016).

[24] P. Contreras-Tejada, C. Palazuelos, and J. I. de Vicente, Genuine Multipartite Nonlocality Is Intrinsic to Quantum Networks, Phys. Rev. Lett. **126**, 040501 (2021).

[25] A. Tavakoli, N. Gisin, and C. Branciard, Bilocal Bell Inequalities Violated by the Quantum Elegant Joint Measurement, Phys. Rev. Lett. **126**, 220401 (2021).

[26] M.-O. Renou and S. Beigi, Network Nonlocality via Rigidity of Token-Counting and Color-Matching, Phys. Rev. Lett. **128**, 060401 (2022).

[27] A. Tavakoli, A. Pozas-Kerstjens, M. X. Luo, and M.-O. Renou, Bell nonlocality in networks, Rep. Prog. Phys. (to be published).

[28] J. Barrett, A. Kent, and S. Pironio, Maximally Nonlocal and Monogamous Quantum Correlations, Phys. Rev. Lett. **97**, 170409 (2006).

[29] L. Masanes, S. Pironio, and A. Acin, Secure device-independent quantum key distribution with causally independent measurement devices, Nat. Commun. **2**, 238 (2011).

[30] L. Aolita, R. Gallego, A. Cabello, and A. Acín, Fully Nonlocal, Monogamous, and Random Genuinely Multipartite Quantum Correlations, Phys. Rev. Lett. **108**, 100401 (2012).

[31] C. M. Lee and M. J. Hoban, Towards Device-Independent Information Processing on General Quantum Networks, Phys. Rev. Lett. **120**, 020504 (2018).

[32] M. X. Luo, Nonsignaling causal hierarchy of general multisource networks, Phys. Rev. A **101**, 062317 (2020).

[33] G. Svetlichny, Distinguishing three-body from two-body nonseparability by a Bell-type inequality, Phys. Rev. D **35**, 3066 (1987).

[34] M.-O. Renou, E. Bäumer, S. Boreiri, N. Brunner, N. Gisin, and S. Beigi, Genuine Quantum Nonlocality in the Triangle Network, Phys. Rev. Lett. **123**, 140401 (2019).

[35] J. Pearl, *Causality* (Cambridge University Press, New York, 2009).

[36] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Proposed Experiment to Test Local Hidden-Variable Theories, Phys. Rev. Lett. **23**, 880 (1969).

[37] T.-C. Wei, I. Affleck, and R. Raussendorf, Affleck-Kennedy-Lieb-Tasaki State on a Honeycomb Lattice is a Universal Quantum Computational Resource, Phys. Rev. Lett. **106**, 070501 (2011).

[38] R. Diestel, *Graph Theory* (Springer-Verlag, Berlin, 2017).

[39] D. M. Greenberger, M. A. Horne, and A. Zeilinger, Going beyond Bell's Theorem, in *Bell's Theorem, Quantum Theory, and Conceptions of the Universe*, edited by M. Kafatos (Kluwer Academic, Dordrecht, 1989), p. 69.

[40] R. F. Werner, Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model, Phys. Rev. A **40**, 4277 (1989).

[41] N. D. Mermin, Extreme Quantum Entanglement in a Superposition of Macroscopically Distinct States, Phys. Rev. Lett. **65**, 1838 (1990).

[42] J. Barrett, R. Colbeck, and A. Kent, Unconditionally secure device-independent quantum key distribution with only two devices, Phys. Rev. A **86**, 062326 (2012).

[43] M. Zukowski, A. Zeilinger, M. A. Horne, and A. K. Ekert, "Event-Ready-Detectors" Bell Experiment via Entanglement Swapping, Phys. Rev. Lett. **71**, 4287 (1993).

[44] M. Navascues, S. Pironio, and A. Acín, Bounding the Set of Quantum Correlations, Phys. Rev. Lett. **98**, 010401 (2007)

[45] S. L. Braunstein and C. M. Caves, Wringing out better Bell inequalities, Ann. Phys. **202**, 22 (1990).

[46] B. F. Toner and D. Bacon, Communication Cost of Simulating Bell Correlations, Phys. Rev. Lett. **91**, 187904 (2003).

[47] C. Branciard, N. Brunner, H. Buhrman, R. Cleve, N. Gisin, S. Portmann, D. Rosset, and M. Szegedy, Classical Simulation of Entanglement Swapping with Bounded Communication, Phys. Rev. Lett. **109**, 100401 (2012).

[48] R. Rabelo, M. Ho, D. Cavalcanti, N. Brunner, and V. Scarani, Device-Independent Certification of Entangled Measurements, Phys. Rev. Lett. **107**, 050502 (2011).

[49] R. Koenig, R. Renner, and C. Schaffner, The operational meaning of min and max-entropy, IEEE Trans. Inf. Theory **24**, 4339 (2009).

[50] R. Raussendorf, D. E. Browne, and H. J. Briegel, Measurement-based quantum computation on cluster states, Phys. Rev. A **68**, 022312 (2003).

[51] I. Affleck, T. Kennedy, E. H. Lieb, and H. Tasaki, Rigorous Results on Valence-Bond Ground States in Antiferromagnets, Phys. Rev. Lett. **59**, 799 (1987).

[52] H. P. Nautrup and T. C. Wei, Symmetry-protected topologically ordered states for universal quantum computation, Phys. Rev. A **92**, 052309 (2015).

[53] A. Childs, Secure assisted quantum computation, Quantum Inf. Comput. **5**, 456 (2005).

[54] A. Broadbent, J. Fitzsimons, and E. Kashefi, Universal blind quantum computation, *Proceedings of the 50th IEEE Symposium on the Foundations of Computer Science (FOCS)* (IEEE, NW Washington, USA, 2009), pp. 517–526.

[55] B. W. Reichardt, F. Unger, and U. Vazirani, Classical command of quantum systems, Nature (London) **496**, 456 (2013).

[56] M. Hajdušek, C. A. Perez-Delgado, and J. F. Fitzsimons, Device-independent verifiable blind quantum computation, arXiv:1502.02563.

[57] T. Morimae and K. Fujii, Secure Entanglement Distillation for Double-Server Blind Quantum Computation, Phys. Rev. Lett. **111**, 020502 (2013).

[58] M. Zwerger, W. Dür, J. D. Bancal, and P. Sekatski, Device-Independent Detection of Genuine Multipartite Entanglement for All Pure States, Phys. Rev. Lett. **122**, 060502 (2019).

[59] N. Gisin and A. Peres, Maximal violation of Bell's inequality for arbitrarily large spin, Phys. Lett. A **162**, 15 (1992).

[60] A. Acín, S. Massar, and S. Pironio, Randomness versus Nonlocality and Entanglement, Phys. Rev. Lett. **108**, 100402 (2012).

[61] A. Coladangelo, K. T. Goh, and V. Scarani, All pure bipartite entangled states can be self-tested, Nat. Commun. **8**, 15485 (2017).

[62] M. Gachechiladze, C. Budroni, and O. Gühne, Extreme Violation of Local Realism in Quantum Hypergraph States, Phys. Rev. Lett. **116**, 070401 (2016).

[63] A. Shamir, How to share a secret, Commun. ACM **22**, 612 (1979).

[64] M. Hillery, V. Buzek, and A. Berthiaume, Quantum secret sharing, Phys. Rev. A **59**, 1829 (1999).

[65] M. G. M. Moreno, S. Brito, R. V. Nery, and R. Chaves, Device-independent secret sharing and a stronger form of Bell nonlocality, Phys. Rev. A **101**, 052339 (2020).

[66] I. Ekeland and R. Temam, *Convex Analysis and Variational Problems*, Vol. 28 (SIAM, Philadelphia, USA, 1999).

[67] J. W. Hall, Local Deterministic Model of Singlet State Correlations Based on Relaxing Measurement Independence, Phys. Rev. Lett. **105**, 250404 (2010).

[68] G. Pütz, D. Rosset, T. J. Barnea, Y. C. Liang, and N. Gisin, Arbitrarily Small Amount of Measurement Independence Is Sufficient to Manifest Quantum Nonlocality, Phys. Rev. Lett. **113**, 190402 (2014).

[69] J. Gallicchio, A. S. Friedman, and D. I. Kaiser, Testing Bell's Inequality with Cosmic Photons: Closing the Setting-Independence Loophole, Phys. Rev. Lett. **112**, 110405 (2014).

[70] J. Handsteiner, A. S. Friedman, D. Rauch, J. Gallicchio, B. Liu, H. Hosp, J. Kofler, D. Bricher, M. Fink, C. Leung *et al.*, Cosmic Bell Test: Measurement Settings from Milky Way Stars, Phys. Rev. Lett. **118**, 060401 (2017).